STUDY GUIDE · PREPARATION FOR TESTS AND THE EXAM

Introduction

to engineering

Weeks 5–7 All savollar, topshiriqlar, atamalar and formulas from the lectures, labs and course templates — with model answers and solutions to the assignments of each week: Problem Framing II and Quantitative Reasoning I–II. Course Introduction to Engineering ·CC-INEN010104 ·4 ECTS Coordinators: Dr Mohammad Reza Bahrami ·Dr Khamidov Ibrokhimjon Fall semester 2026$SAVOLLAR \cdot TOPSHIRIQLAR \cdot ATAMALAR \cdot FORMULALAR \cdot QOIDALAROCTOBER2026$

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1 1. Course map, sources, and how to use this guide

This study guide continues the series of guides for the course Introduction to Engineering (code CC-INEN010104, 4 ECTS, fall semester 2026/27, Tashkent University of Technology, Faculty of Power Engineering) and covers weeks 5–7: Problem Framing and Modelling II (week 5), Quantitative Reasoning I (week 6) and Quantitative Reasoning II (week 7). Every savol (savollar

- questions), topshiriq (topshiriqlar — assignments), atama (atamalar — terms), formula (formulalar) and qoida (qoidalar — rules) of these weeks, encountered in the lectures, labs and templates, is written out in full and supplied with a model answer or solution. The guide is structured so that you can prepare for the weekly mini projects (30%), the midterm $(30\%) and$ the final (40%) without re-watching hundreds of slides.

The course is taught by Dr. Mohammad Reza Bahrami (lectures; Faculty of Power Engineering, PhD Mechanical Engineering) and Ibrokhimjon Khamidov (labs; lab instructor; research interests — lean manufacturing, sustainability, quantitative assessment of productivity). The course follows a Project-Based Learning approach: all semester work revolves around a single team system selected in week 2. Week 5 ends with a team deliverable, Milestone 2 — a problem-framing document and an explicit list of modelling assumptions; labs 5–7 keep working with the same ten scenario cards as Lab 4. Lectures 5–7 draw on P. Kosky, R. Balmer, W. Keat, G. Wise, "Exploring Engineering" (5th ed.): chapter 2 — the Need-Know-How-Solve method, units in square brackets, significant figures and order-of-magnitude estimates (Examples 2.3–2.4, Problem 10); chapter 4 — models and "beware of your assumptions" (Example 4.8); chapter 8

- highway capacity as an intensity; chapter 10 — the electric circuit model.

How to use this guide: sections 2–4 follow the week-by-week chronology; within each week there is first a digest of "all the qoidalar of the week" (what gets tested first of all), then all the savollar of the lecture and the lab with model answers, then the lab's topshiriqlar with solutions and a glossary of atamalar. Sections 5–8 are summary sections: all the formulas (Formulalar), all the rules (Qoidalar), all the assignments (Topshiriqlar) and a cheat sheet of key facts. The terminology is kept bilingual (Russian + English) because the exam is administered in English; the numbers in all examples are illustrative values from the course slides, not measured data. Week 8 is the Midterm Assessment: a written, paper-based test covering the first seven weeks.

1.1 1.1. Weeks 5–7 and the course files

Week	Topic	Lecture	Laboratory	Templates
5	Problem framing and modelling II: models as purposeful simplifications; four types and four parts of a model; assumptions; fit for purpose	Lecture05 ProblemFraming andModelling II.txt (33 slides)	Lab5Problem FramingII.txt (16 pages)	Engineering ProblemFraming Blanket Template.txt; Lab5Model Worksheet Template.txt
6	Quantitative reasoning I: demand and capacity, utilisation, peaks, intensities and flows, order of magnitude	Lecture06 Quantitative ReasoningI.txt (32 slides)	Lab6Quantitative ReasoningI.txt (17 pages)	Lab6Worksheet Template.txt
7	Quantitative reasoning II: incomplete data, ranges, triangulation, interpretations, honest reporting	Lecture07 Quantitative ReasoningII.txt (32 slides)	Lab7Quantitative ReasoningII.txt (21 pages)	Lab7Engineering Decisions IncompleteData Template.txt

Table 1.1. Weeks 5–7 and the course files on which this guide is based

[Source: text extracts of the course files course-content/ite; the lectures are delivered by Dr. Mohammad Reza Bahrami,

the labs are run by Ibrokhimjon Khamidov, Fall 2026]

2 2. Week 5 — Problem framing and

modelling II (Problem Framing and Modelling II)

2.1 2.1. Key statements (“all the qoidalar” of week 5)

The week 5 lecture turns the framed problem into a simple model and teaches you to state what it assumes. A recap of week 4 “in one breath”: sort the evidence into fact, inference, speculation and opinion; needs, not solutions — a good problem statement does not name a device; variables are measurable quantities with units (performance, design, context); performance is defined by the stakeholders — write it down in numbers.

A model is a purposeful simplification

A model is a simplified representation of a real system, built to answer a specific question. Every model keeps something in (for the question of a bus’s range — battery energy, distance, consumption per kilometre), throws something away (the colour of the bus, the number of seats, the exact shape of every turn) and always has a purpose — a question: “is one charge enough for a day on the route?”.

The lecture’s opening example is one campus electrical network and two maps. The one-line diagram (Map A) answers connectivity questions: which buildings lose

power if the library feeder is switched off; where to place a switch; which switch isolates a fault; it omits distances, cable cross-sections and time. The geographic diagram (Map B) answers location questions: how long the cable to the workshop is, which lines cross the river or the road, which building is closer to the substation; it omits switching, protection and power ratings. Neither map is the network itself: each is a model, a simplified picture for one type of question. The week's epigraph is the statistician George Box: "All models are wrong, but some are useful."

[Source: Lecture05, slides 2, 4–7]

Four types of models

The course distinguishes four types. Physical — a real copy: a scaled-down or simplified object, for example a wind turbine blade tested in a wind tunnel. Graphical — a diagram: a drawing of parts and connections, for example a one-line diagram or the energy-flow diagram of a car. Mathematical — equations: relations between variables, for example Ohm's law and the power law for an electric circuit. Computational — a calculation: rules executed step by step on a computer, for example a table of hourly electricity demand. Real projects mix several types: a diagram to organise, equations to compute, a table

- to run the numbers.

One system — three models (campus electricity supply): the one-line diagram (graphical; answers "which buildings lose power when a feeder is switched off?"; omits distances, power ratings, time); the energy balance supply = demand + losses (mathematical; answers "does daily generation cover daily demand?"; omits the hour in which demand arises); the hourly demand table (computational; answers "in which hour does the peak occur?"; omits the network diagram and the voltages). One system, three models — the question decides which one to use.

[Source: Lecture05, slides 8–9; Lab5, p. 3]

Anatomy of a model: four parts

Every model consists of four parts. Inputs — what we feed in: the values we choose or can change (battery size, speed, route length). Parameters

- — what is kept fixed: the numbers describing the system and its conditions (energy consumption per kilometre, the usable share of the battery). Relationships — the rules: equations or steps that link the numbers (Energy = energy per km $\times$ distance). Outputs — what the model predicts: the results we evaluate (range in km, energy in kWh/day). "A model

- is a machine: numbers go in, a rule is applied, numbers come out" (A model is a machine). Every number needs its own units.

The week 4 variables become the roles of the model: performance variables $\rightarrow$ outputs (range per charge [km], energy per day [kWh/day]); design variables $\rightarrow$ inputs (battery capacity [kWh], number of buses); context variables $\rightarrow$ parameters (route length [km], air temperature [°C], hours on depot charging). Last week's list of variables is already the skeleton of this week's model.

[Source: Lecture05, slides 13–14; Lab5, p. 4]

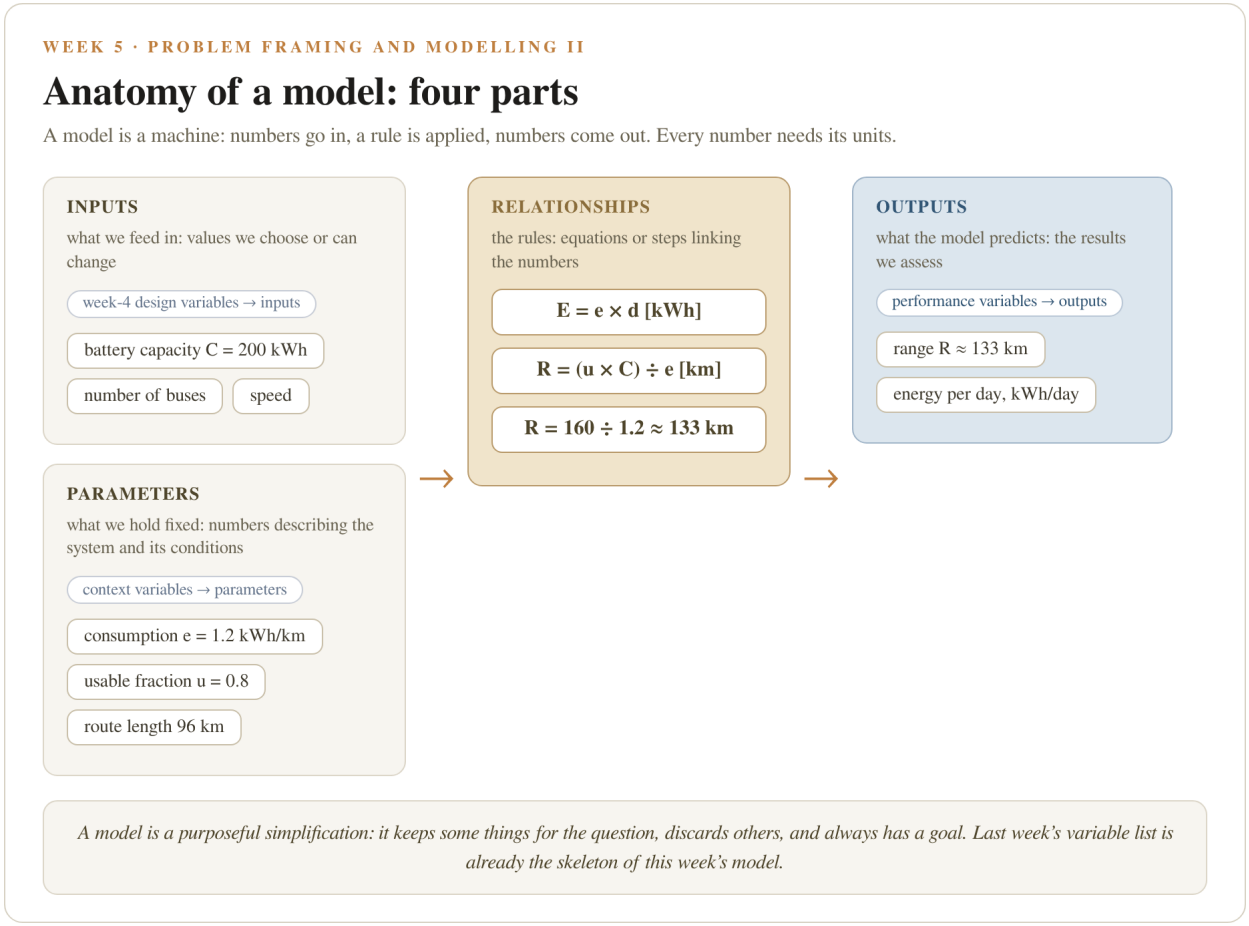


Fig. 1. Anatomy of a model (week 5): inputs, parameters, relationships and outputs — the four parts of a model, with week-4 variables in their model roles.

Need–Know–How–Solve and the first bus model

The week's method is Need–Know–How–Solve (Kosky, ch. 2): Need — the variable we want (range per charge, km); Know — what is already known (battery capacity, usable share, consumption per km, route length); How — the model and the assumptions (Range = usable energy ÷ energy per km); Solve — compute, check the units, interpret. Kosky stresses that How also includes the assumptions you make in order to solve the problem. Week 4 gave the Need; week 5 builds the How — and the assumptions hidden inside it.

The first model is an electric campus bus: trip energy $E = e \times d$, range $R = (u \times C)$ ÷ e, where C is the battery capacity [kWh], u is the usable share (unitless), e is the consumption [kWh/km], d

- is the distance [km], R [km], E [kWh]. A unit check after Kosky: $(u \times C)$ [kWh] ÷ e [kWh/km] = R [km].

$E = e \times d$ [kWh] = [kWh/km] × [km]

$R = (u \times C) \div e \rightarrow 160$ [kWh] ÷ 1,2 [kWh/km] = 133 [km]

[Source: Lecture05, slides 15–16; method: Kosky, ch. 2]

The model's numbers: 160 kWh, 133 km, 96 km

Substituting the numbers (illustrative, not data): usable energy $u \times C = 0,8 \times 200$ kWh $= 160$ kWh; range $R = 160$ kWh ÷ 1,2 kWh/km $= 133$ km; the daily requirement is 8 laps $\times 12km =$

96 km. The model says: one charge covers the day with a margin of about 37 km. But how far can this number be trusted? That depends on the assumptions.

[Source: Lecture05, slide 17]

Three kinds of assumptions and a template for writing them

Simplifying — what we ignore: hills, traffic lights, stops, differences between drivers. Data — what we fix: $e = 1,2$ kWh/km; $C = 200$ kWh, $u = 0,8$. Boundary — what we leave outside: the passengers, the charger and the grid; only the bus is in the model. The idea of week 2 repeats: the boundary is a choice, and every assumption is a choice too.

The course template: “We assume [assumption] because [reason]. If it is wrong, [effect on the result]”. Example: “We assume 1,2 kWh/km in all seasons, because we only have autumn test data. If it is wrong, the winter range may fall below the daily requirement of 96 km”. A quick check: will a teammate be able to understand what you ignored, fixed and left outside? Can it be checked? An assumption that is written down can be checked.

A hidden one cannot.

We assume [assumption] because [reason]. If it is wrong, [effect on the result].

[Source: Lecture05, slides 19, 23; Lab5, p. 5]

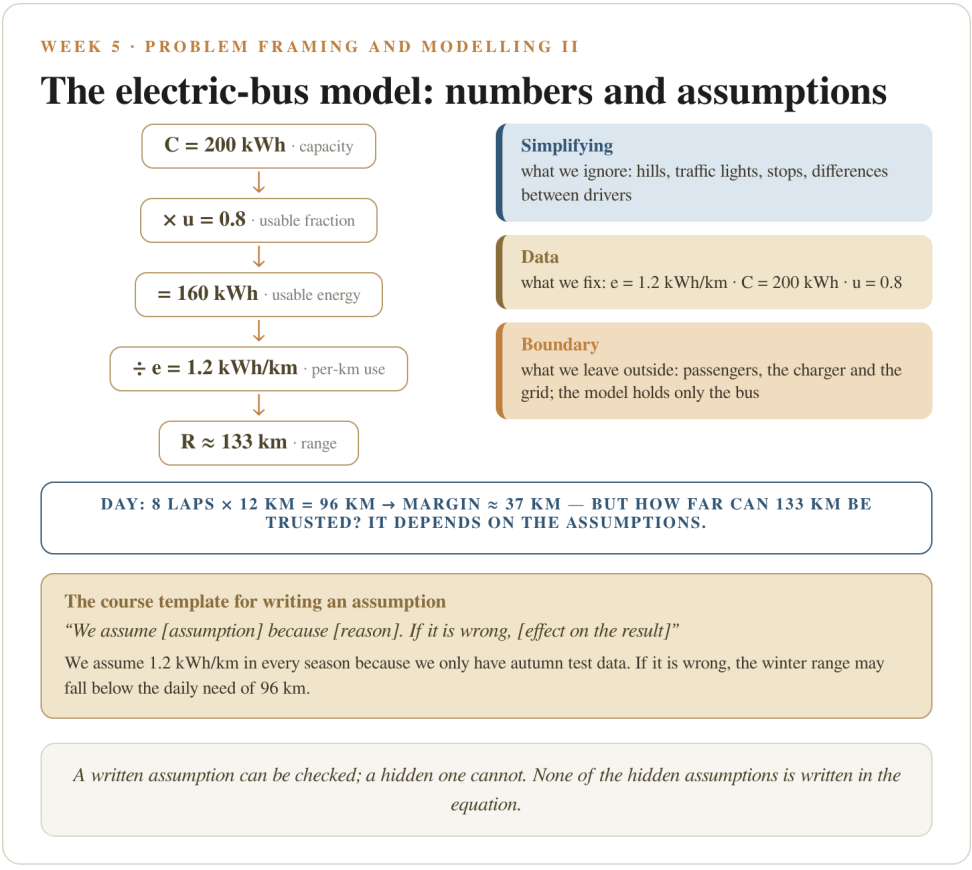


Fig. 2. The electric-bus model (week 5): the numbers 160 kWh / 133 km / 96 km, the three kinds of assumptions and the writing template.

Hidden assumptions and their price

None of the hidden assumptions of the bus model is written in the equation. $e = 1,2$ kWh/km — "the same in every season": cabin heating raises e, and the range drops. "The route is flat, the load is similar": hills and full buses spend more energy. $u \times C = 160$ kWh — "the battery never loses capacity": an ageing battery stores less. "We start from a full battery": depot charging always works; a missed charge cuts the day short. This is exactly why assumptions are written down.

One model — different assumptions: base case 133 km; full passenger load 114 km; old battery 107 km; winter heating 89 km; winter + old battery 71 km (cases: $e = 1,4 and 1,8$ kWh/km, usable energy 128 kWh). The daily requirement is 96 km: the base case, the full load and the old battery cover the day; winter heating and "winter + old battery" no longer do. Conclusion: "one charge covers the day" holds only for as long as the assumptions hold.

A solar example of the same idea: the hidden assumption "the sun shines at full power all day". The naive model: 5 kW $\times 24h = 120$ kWh per day; the clear-day profile is about 31 kWh per day. The naive model is nearly four times too high. Note the units: kW is power, kWh is energy. Before trusting a number, ask what conditions it silently assumes.

[Source: Lecture05, slides 20–22]

Is the model good enough? (Fit for purpose)

Three verdicts. Too simple (too simple) — it misses what matters: $R = 160$ ÷ 1,2 for every day of the year; winter, hills and ageing are invisible. Fit for purpose (fit for purpose) — it answers the question: separate consumption for summer and winter plus a safety margin; enough to decide whether one charge covers the day. Too detailed (too detailed) — it cannot be built and checked: battery chemistry, every traffic light, every driver's habits; it needs data that do not exist. The rule:

choose the simplest model that answers the question well enough to make a decision.

When a model misleads: the range model $(e = 1,2 kWh/km)$ was built for mild autumn test days but is used during a winter cold snap — heating raises e, and the range is overestimated; the solar generation profile was built for a clear summer day but is used in a cloudy winter week — generation falls far below the profile; the dormitory load of 2 kW is for mild weather but is used in a week of hard frost — the load is far above 2 kW. Every model has a range within which its assumptions are valid. Know it and state it.

[Source: Lecture05, slides 27, 29]

Three checks and the "build — check — revise" loop

Three sanity checks: (1) units — $(u \times C)$ [kWh] ÷ e [kWh/km] gives km, so the range has the right unit; (2) order of magnitude — are 133 km plausible for a city bus: tens–hundreds of km, not 3 km and not 3 000 km; (3) a known value — does the result agree with a datasheet, a test drive or last month's logs. Kosky ends the energy example with the question "Is this a reasonable answer?" — ask it every time.

The week's cycle: Frame (the problem statement and the variables) $\rightarrow$ Build (a first simple model with units) $\rightarrow$ Check (units, magnitude, known values) $\rightarrow$ Revise (fix the assumptions; add detail only if needed) — repeat until it is good enough for the decision. A preview: in week 11 the sensitivity analysis will ask "what if this assumption is wrong — and by how much?"

[Source: Lecture05, slides 28, 30–31]

2.2 2.2. Savollar of week 5 — lecture and lab questions with model answers

S1. What is a model? Why does the lecture insist that "no map is the network"?

Answer. A model is a simplified representation of a real system, built to answer a specific question. The lecture's example: one campus electrical network and two maps — the one-line diagram (connectivity questions: which buildings lose power when a feeder is switched off, where to place a switch, which switch isolates a fault) and the geographic diagram (location questions: the cable length to the workshop, crossings of the river or the road, the building closest to the substation). Neither map is the network: each is a model, a simplified picture made for one type of question. George Box's epigraph: "All models are wrong, but some are useful".

S2. Lecture interactive "Match the Question to the Model": which type of model fits each of the six questions A–F best, and why?

Answer. A (range on a single charge) — mathematical: energy = consumption per km $\times$ distance set against the battery's energy. B (which buildings lose power when a feeder is switched off) — graphical: the one-line diagram shows which feeder supplies a building. C (dormitory temperature over a day) — computational: we step through the hours. D (a turbine blade in a strong wind) — physical: the blade in a wind tunnel. E (where passengers stand at a busy stop) — graphical: a sketch of the stop. F (a building's energy over a day) — mathematical: kW $\times h$ = kWh from the loads. Several types may fit — explain why.

S3. What four parts does every model consist of? Give an example of each for the electric bus.

Answer. Inputs — what we feed in and can change: battery size, speed, route length. Parameters

- — what is kept fixed: consumption per km, the usable share of the battery. Relationships — the rules linking the numbers: Energy = energy per km $\times$ distance. Outputs — the predictions we evaluate: range [km], energy per day [kWh/day]. The lecture's formula: "a model is a machine: numbers go in, a rule is applied, numbers come out", and every number needs its own units.

S4. How do the week 4 variables (performance, design, context) become the roles of the model?

Answer. Performance variables become the outputs: range per charge [km], energy per day [kWh/day]. Design variables become the inputs: battery capacity [kWh], number of buses. Context variables become the parameters: route length [km], outside temperature [°C], hours on depot charging. Last week's list of variables is already the skeleton of this week's model: a variable's role is determined by how it enters the calculation.

S5. Apply Need–Know–How–Solve to the bus range question.

Answer. Need — the variable we want: range per charge [km]. Know — what is known: battery capacity, usable share, consumption per km, route length. How — the model and the assumptions: Range = usable energy ÷ energy per km, i.e. $R = (u \times C)$ ÷ e; after Kosky, How also includes the assumptions made in order to solve the problem. Solve — compute, check the units, interpret: $(u \times C)$ [kWh] ÷ e [kWh/km] = R [km]. Week 4 gave the Need; week 5 builds the How — and the assumptions hidden inside it.

S6. Substitute the numbers into the bus model: what are the usable energy, the range and the daily requirement?

Answer. Usable energy $u \times C = 0,8 \times 200$ kWh $= 160$ kWh. Range $R = 160$ kWh ÷ 1,2 kWh/$km = 133$ km. Daily requirement: 8 laps $\times 12km = 96$ km. The model's conclusion: one charge covers the day with a margin of about 37 km. The lecture immediately warns: how far the number can be trusted depends on the assumptions; the values are illustrative, not measured data.

S7. Which three kinds of assumptions does the course distinguish? Give an example of each for the bus model.

Answer. Simplifying — what we ignore: hills, traffic lights, stops, differences between drivers. Data assumptions — what we fix: $e = 1,2$ kWh/km; $C = 200$ kWh, $u = 0,8$. Boundary — what we leave outside the model: the passengers, the charger, the grid; only the bus is in the model. Every assumption is a choice (like the system boundary from week 2), and each is written down with the template: "We assume … because …. If it is wrong, …".

S8. Lecture exercise "Spot the Hidden Assumptions": find the hidden assumption in the statement "The dormitory needs a 2 kW heater, so a day costs 48 kWh" and rewrite it with the template.

Answer. The hidden assumption: the heater runs at full power for all 24 hours, regardless of the weather. Rewrite: "We assume the heater is always on, because we have no thermostat data. If it switches off in cycles, the energy is below 48 kWh". Similarly: "3 000 kWh per day $\rightarrow 90000$ kWh over 30 days" — the assumption "every day is an average day": if winter days consume more, the month will exceed 90 000. "1,2 kWh/$km \rightarrow$ ten buses over a 12 km lap each $= 144$ kWh" — the assumptions "consumption is the same for all" and "one lap each": if some laps are full, the total will exceed 144 kWh.

S9. How does the range change under full passenger load, an old battery and winter heating?

Answer. Base case — 133 km; full passenger load — 114 km; old battery — 107 km; winter heating — 89 km; winter + old battery — 71 km (cases: $e = 1,4 and 1,8$ kWh/km, usable energy 128 kWh). The daily requirement is 96 km: the first three cases cover the day; winter heating and "winter + old battery" do not. The lecture's moral: "one charge covers the day" holds only for as long as the assumptions hold.

S10. Why is the naive estimate of a solar array, 5 kW $\times 24h = 120$ kWh/day, nearly four times too high?

Answer. The hidden assumption is "the sun shines at full power all day". The naive model gives 5 kW $\times 24h = 120$ kWh per day; the clear-day profile is about 31 kWh per day, i.e. nearly four times less. The lesson: before trusting a number, ask what conditions it silently assumes. And keep the units apart: kW is power, kWh is energy.

S11. What does "fit for purpose" mean, and what are the two opposite ways a model can fail?

Answer. Fit for purpose — the model is good enough to answer the question and make a decision. The "too simple" failure: $R = 160$ ÷ 1,2 for every day of the year — winter, hills and ageing are invisible. The "too detailed" failure: battery chemistry, every traffic light, every driver's habits — the model cannot be built or checked, and needs inaccessible data. A fit-for-purpose model — for example, separate consumption for summer and winter plus a safety margin — is enough to decide whether one charge covers the day. The rule: choose the simplest model that answers the question well enough to make a decision.

S12. Which three sanity checks does a model pass, and when does a model "mislead"?

Answer. The checks: (1) units — $(u \times C)$ [kWh] ÷ e [kWh/km] gives km; (2) order of magnitude — 133 km is plausible for a city bus (tens–hundreds of km, not 3 and not 3 000); (3) a known value — a datasheet, a test drive, last month's logs. A model misleads outside the range of its assumptions: the autumn range model — in a winter cold snap; the clear summer day profile — in a cloudy winter week; the 2 kW load for mild weather — in a frosty week. Every model has a range in which its assumptions are valid: know it and state it.

S13. Lab 5: what does the assessed team work consist of (ten steps), and how is it graded?

Answer. The team takes the same scenario card as in Lab 4 and, in 45 minutes, works through ten steps: (1) the model's question; (2) the type, with justification; (3) the anatomy — inputs, parameters, relationships, outputs, all with units; (4) Need–Know–How–Solve; (5) at least six assumptions using the template; (6) labelling Safe / Risky / Need Data; (7) changing one assumption and the effect on the output; (8) the checks — units, order of magnitude, a known value; (9) the verdict Too Simple / Fit for Purpose / Too Detailed; (10) a revision and the need for data. Grading per step: 3/2/1/0 (fully / partially / almost nothing / missing); the steps go in order — each next one depends on the previous ones.

S14. Lab 5, model answer "EV Charging & Battery Test Station": what are the model's question, type, anatomy and result? What happens on switching to 50 kW fast charging?

Answer. The question: will the station's peak power stay within the grid operator's 150 kW limit in the busiest hour? Type — mathematical. Anatomy: inputs — the chargers in use n [units], the power of one charger P_1 [kW]; parameters — the limit $L = 150$ kW, 4 chargers; the relationship $P = n \times P_1$, compliant if $P \le L$; outputs — P [kW], yes/no. Solve: $P = 4 \times 22 = 88$ kW ≤ 150 kW. A stress test: at $P_1 = 50$ kW, $P = 4 \times 50 = 200$ kW — above the limit; the model exposes a hidden violation. Verdict: Fit for Purpose; the revision — a measured diversity factor instead of the worst case; the data item — a week of measurements per charger.

[Source: Lecture05; Lab5 — lecture and lab questions of week 5]

2.3 2.3. Topshiriqlar of the week 5 lab — with solutions

Lab 5 lasts 80 minutes: a recap of lecture $5 (15\,\text{min})$ — what a model is, the four types, the anatomy and assumptions; returning the scenario cards $(5\,\text{min})$ — the same ten cards from Lab 4 / Homework, the team keeps its own; the team work "Build & Check a Model" $(45\,\text{min}$, assessed) — fill in the model template: anatomy, Need–Know–How–Solve, assumptions, checks, fit for purpose; the model answer walkthrough $(15\,\text{min})$ — a worked model for the EV Charging & Battery Test Station scenario.

[Source: Lab5, p. 2]

Assignment L5-1. Build & Check a Model — ten steps (assessed, 45 minutes)

Statement: use the same scenario card as in Lab 4 / Homework and build a working model for it, with stated assumptions. Ten steps:

- Define one clear question/purpose of the model (write it as a question).
- Choose the type of model and justify it.

- Build the anatomy: inputs, parameters, relationships, outputs — all with units.
- Fill in Need–Know–How–Solve.
- Write out at least 6 assumptions: simplifying, data, boundary — using the template.
- Label each assumption Safe / Risky / Need Data.
- Change one assumption; explain the effect on the output.
- Check the model: units, order of magnitude, a known value.
- Rate the model: Too Simple / Fit for Purpose / Too Detailed.
- Name one revision and one additional need for data.

Grading per step: 3 — good/fully done; 2 — partially; 1 — present, almost nothing done; 0 — missing. The work is graded live on all ten steps; do them in order — each next step depends on what is written in the previous ones.
[Source: Lab5, p. 7]

Model answer L5-1 — the "EV Charging & Battery Test Station" scenario

Step 1. The model's question. "Will the station's peak power draw stay within the grid operator's 150 kW limit during the busiest hour?"

Step 2. The type of model is mathematical. Chosen because the question is a direct calculation: the power per charger $\times$ the number of chargers in use, compared against the limit.

Step 3. The anatomy of the model.

Part	Name (with unit)	Role
Inputs	chargers in use n [units]; power per charger P_1 [kW]	what we feed in / can choose
Parameters	grid limit $L = 150$ [kW]; number of chargers $= 4$	what we keep fixed
Relationships	total power: $P = n \times P_1$; compliant if $P \le L$	equation / rule
Outputs	total power P [kW]; compliance (yes/no)	what the model predicts

Table 2.1. Anatomy of the charging-station model (Lab 5, step 3)

Step 4. Need–Know–How–Solve. Need — the peak power [kW]. Know — $P_1 = 22$ kW per charger; 4 chargers; the 150 kW limit. How — $P = n \times P_1$, with $n = 4$ *at* the peak. Solve — $P = 4 \times 22 = 88$ kW ≤ 150 kW;

- compliance is confirmed.

Steps 5–6. Six assumptions from the template and their classification.

We assume …, because …	Kind	Risk	If wrong …
All 4 chargers draw 22 kW simultaneously at the peak, because that is the rated power of each unit	data	Safe	the actual peak power will be lower, not higher
Charging power is constant over a session, because we ignore the power taper near full charge	simplifying	Risky	the real peak is slightly below the modelled one
The diagnostic test consumes a negligible amount, because it is a low-current measurement	simplifying	Safe	the model slightly understates consumption
Only the 4 charging bays are inside the model boundary, because the building's lighting/HVAC is metered separately	boundary	Need Data	the building load may add to the connection
Battery condition checks add no electrical load, because they use the charger's own diagnostics	boundary	Safe	no change to the result is expected
All 4 chargers can be occupied simultaneously in the peak hour, because this is the worst case, not the average	data	Risky	if fewer than 4 are occupied at once, consumption is lower

Table 2.2. The model answer's assumptions and their labelling (Lab 5, steps 5–6). Safe — holds under normal conditions; Risky — may fail and change the answer; Need Data — cannot be said yet

Step 7. Changing one assumption. The chargers are switched to fast charging: P_1 rises from 22 to 50 kW per charger. The new result: $P = 4 \times 50 = 200$ kW — above the 150 kW limit. The model exposes a compliance violation that the original assumption was hiding.

Step 8. Three checks of the model.

Check	Result
Units	n [units] $\times P_1$ [kW] $= P$ [kW] — the units are consistent
Order of magnitude	88 kW for 4 fast chargers is plausible; not 8 kW (too small) and not 8 000 kW (too large)
A known value	matches the datasheets of public DC chargers (~20–25 kW per standard unit)

Table 2.3. Sanity checks for the station model (Lab 5, step 8)

Steps 9–10. Verdict, revision, need for data. Verdict: Fit for Purpose — the model answers the compliance question (yes/no against 150 kW) with the data available, without simulating the minute-by-minute load curve. Revision: replace the worst case "all 4 occupied at once" with a measured diversity factor from real usage logs. Additional need for data: a week of measured power per charger, to confirm the 22 kW rating in practice.

[Source: Lab5, pp. 11–14]

Assignment L5-2. Model Building & Assumption Testing Worksheet (the week 5 sheet)

The homework sheet repeats the ten steps of the lab. The sheet's instruction: "Use the same scenario card as in Lab 4 / Homework. Turn it into a working model: define its question and type, build the anatomy, apply Need–Know–How–Solve, write out and classify the assumptions, stress one of them, sanity-check the result and decide whether the model is fit for purpose". The sheet's items:

- Model Question / Purpose — the one question the model must answer; write it as a question.
- Model Type & Justification — Physical / Graphical / Mathematical / Computational, and why this type

fits.

- Model Anatomy — name each part with its unit: Inputs (what we feed in / can choose),

Parameters (what we keep fixed), Relationships (an equation or a rule), Outputs (what it predicts).

- Need–Know–How–Solve — the four cells of the table.
- At least 6 assumptions — simplifying, data and boundary; each in the lines: "We assume ...

because .... If it is wrong, ...".

- Classification of each assumption: kind (simplifying / data / boundary) and Safe / Risky / Need Data.
- Changing one assumption: the old value, the new value, the effect on the output.
- Checking the model: units — do they agree; order of magnitude — is the number plausible;

a known value — does it match a datasheet or a known case.

- Verdict: Too Simple / Fit for Purpose / Too Detailed — and why.
- One revision and one additional need for data.

The sheet's model solution — the model answer of assignment L5-1 (Tables 2.1–2.3 and steps 4, 7, 9–10).
[Source: Lab5ModelWorksheetTemplate.txt]

Assignment L5-3. Engineering Problem Framing — the cross-cutting template (30–40 minutes, team work)

The team worksheet on problem framing. Instruction: "Read the scenario carefully. Use the textual evidence to classify the statements, frame the problem, draw a process-flow diagram, define the variables with units, lay out the stakeholders' views

and propose must-have and nice-to-have goals. The problem statement must be free of solutions". Below are sections A–H with model answers for the EV Charging & Battery Test Station card.

A. Sort the Evidence. Classify the statements of the scenario: two each of fact, inference, speculation and opinion, each with an explanation of "why this category". Model examples for the station card: fact — the grid operator's limit $L = 150$ kW; fact — each of the 4 charging bays is rated at 22 kW; inference — with four chargers running simultaneously, the peak power is 88 kW ≤ 150 kW (a model calculation); inference — if fewer than four are occupied at once, consumption is below 88 kW; speculation — switching the chargers to fast mode will raise P_1 to 50 kW (the stress-test hypothesis of step 7); speculation — the building's lighting/HVAC will add load on the same connection (a boundary assumption, Need Data); opinion — "the model is good enough: it answers the compliance question without simulating the load curve" (the verdict of step 9); opinion — "replace the worst case with a measured diversity factor" (the revision recommendation of step 10).

[The evidence is taken from the Lab 5 model answer; using the template, the team writes its own $2 + 2 + 2 + 2$ from its own

card]

B. Complaint $\rightarrow$ Need $\rightarrow$ Problem Statement. The path: complaint (what people say) $\rightarrow$ need (what they actually require) $\rightarrow a$ problem statement without a solution. The statement template: [Stakeholder] needs [outcome] in [system], judged by [measure], within [constraints]. Model answer: the station operator needs to keep the peak power within the grid operator's limit in the busiest hour; the statement: "The station needs to keep its peak power draw within the grid operator's limit, judged by total peak power P [kW] against $L = 150$ kW, within the four-bay charger configuration" — without naming specific solution devices.

C. Process Flow Diagram (6–8 main steps). Use the symbols Start/End, Step, Decision (Yes/No), Wait/Queue and Flow arrows; label what moves between the steps and mark the variables/units. A model for the station (from the actions named in the lab's model answer): Start $\rightarrow$ arrival of an electric vehicle $\rightarrow$ connection to a charging bay ($n + 1$ occupied [units]) $\rightarrow$ charging session ($P_1 = 22kW$, constant) $\rightarrow$ the decision "is a diagnostic test needed?" $\rightarrow a$ low-current diagnostic test (negligible power) $\rightarrow$ battery condition check via the charger's own diagnostics (no additional load) $\rightarrow thebay$ is freed $(n - 1) \rightarrow$ End; outside the boundary — lighting/HVAC, metered separately.

D. Variables on the Process. At least 5 variables, including at least one performance, one design and one context; for each — the step/flow, the unit, the type (P / D / C) and how to measure it. Model: performance

- — total peak power P [kW] (from the meter) and compliance with the limit [yes/no]; design — the number of charging bays [units], the charger power P_1 [kW]; context — the grid limit $L = 150$ kW, the lighting/HVAC load [kW] (outside the boundary), the hour of peak occupancy [h of the day].

E. Stakeholder Performance View. The table "stakeholder $\rightarrow$ their idea of success $\rightarrow$ candidate variable $\rightarrow$ unit". Model row: grid operator — success: the peak power stays within 150 kW — the variable P [kW]. The remaining rows are filled in from the stakeholders of the team's card.

F. Must-have and Nice-to-have Targets. Rough targets for at least 3 variables: must-have = demand, nice-to-have = wish. Model: peak power $P \leq 150$ kW in the busiest hour (must-have — the operator's limit); $P \leq 150$ kW even after the switch to fast charging at $P_1 = 50$ kW (nice-to-have: today $P = 4 \times 50 = 200$ kW — the goal is not met

without load management); a measurement-confirmed rating of 22 kW per charger (nice-to-have — the data need of step 10).

G. Final Working Definition of System Performance. Complete the sentence: “The performance of ___ is judged by ___, ___ and ___, each with units and a target, through the eyes of ___”. Model: the performance of the EV Charging & Battery Test Station is judged by the total peak power P [kW] and compliance with the 150 kW limit (yes/no), with units and targets, through the eyes of the grid operator.

H. 60-Second Team Report-Out (optional). Four questions: what is your solution-free problem statement? which 2–3 variables and units matter most? which stakeholder matters most in your statement, and why? which one must-have goal and one nice-to-have goal?

[Source: EngineeringProblemFramingBlanketTemplate.txt; the model answers are for the EV Charging & Battery Test

Station card from Lab 5]

Assignment L5-4. Milestone 2 — Independent Study (after lecture 5)

- Write a problem-framing document: the problem statement, the performance variables,

the demands and wishes.

- Draw the model as a block: inputs, parameters, relationships, outputs — each with units.

- Write out at least eight assumptions: what we assume, why, and what changes if it is

wrong.

- Label each assumption safe, risky or needs data.

The team deliverable: Milestone 2 — a problem-framing document and a list of modelling assumptions. Next week the course starts putting numbers into the model.

[Source: Lecture05, slide 32]

2.4 2.4. Atamalar — glossary of week 5

Term (RU)	English	One-phrase definition
Model	Model	A simplified representation of a real system, built to answer a specific question
Physical model	Physical model	A real scaled-down or simplified copy: a turbine blade in a wind tunnel
Graphical model	Graphical model	A diagram of parts and connections: a one-line diagram, an energy-flow diagram
Mathematical model	Mathematical model	Equations between variables: Ohm's law, an energy balance
Computational model	Computational model	Rules executed step by step on a computer: a table of hourly demand
Inputs	Inputs	What we feed into the model and can choose: battery size, speed, route length
Parameters	Parameters	What we keep fixed: consumption per km, the usable share of the battery
Relationships	Relationships	The rules linking the model's numbers: Energy = energy per km × distance
Outputs	Outputs	The predictions we evaluate: range [km], energy [kWh/day]
Simplifying assumption	Simplifying assumption	What we ignore: hills, traffic lights, differences between drivers
Data assumption	Data assumption	What we fix: $e = 1,2$ kWh/km; $C = 200$ kWh, $u = 0,8$
Boundary assumption	Boundary assumption	What we leave outside the model: passengers, charging, the grid
Fit for purpose	Fit for purpose	The simplest model that answers the question well enough for a decision
Sanity check	Sanity check	Units, order of magnitude, comparison with a known value

[Source: Lecture05; Lab5]

3 3. Week 6 — Quantitative Reasoning I

3.1 3.1. Key statements ("all the qoidalar" of week 6)

The week 6 lecture puts numbers on demand, capacity, intensities and flows, and teaches you to check them by estimation. A recap of week 5 "in one breath": models are purposeful simplified representations; four parts — inputs, parameters, relationships, outputs, all with units; assumptions — what we ignore, fix or leave outside, and we write them down; fit for purpose — the simplest model that answers the question well enough to decide.

Demand and capacity: what is asked, and what the system can do

Demand is what the system must deliver. Capacity is the maximum it can deliver. Both need units and a time basis: passengers per hour, kW at 13:00, kWh per day. A number without a time basis is as incomplete as a number without units. The idea from Kosky, ch. 8: highway capacity is an intensity.

System	Demand	Capacity	Unit
Campus bus at the peak	300 passengers want to ride	240 can be carried (40 seats $\times 6$ buses)	per hour
Canteen at lunch	360 diners arrive	400 meals can be served	per hour
Depot connection	360 kW if six chargers run	200 kW provided by the connection	kW
Faculty building at the peak	180 kW consumed at 13:00	250 kW transformer rating	kW

Table 3.1. Demand and capacity in the lecture's four systems (illustrative values)

Always compare like with like: one unit and one time basis on both sides.

[Source: Lecture06, slides 6–7; Lab6, p. 3]

Utilisation: demand ÷ capacity

Utilisation = demand ÷ capacity. The faculty building: 180 ÷ 250 kW $= 0,72$ — comfortable. The canteen: 360 ÷ 400 per hour $= 0,90$ — tight. The campus bus: 300 ÷ 240 per hour $= 1,25$ — overload. The depot, all six chargers: 360 ÷ 200 kW $= 1,80$ — overload. Empirical thresholds: below ~{}0,8 — comfortable (there is margin to absorb variation); ~{}0,8–1,0 — tight (little margin, delays appear); above 1,0 — overload: demand is turned away or a queue grows.

Utilisation = Demand ÷ Capacity (the same units, the same time basis)

[Source: Lecture06, slide 8; Lab6, p. 3]

The average hides the peak: size for the peak

The load profile of the faculty building: an average of 94 kW, a peak of 180 kW at 13:00, a transformer capacity of 250 kW. A system sized for the average would fail at midday: "Sized for the average, the building would fail at midday". The week's rule: Size for the peak, not for the average. The average hides the peak — which is why 86 kW of average charging power does not mean that six 60-kW chargers can run simultaneously.

[Source: Lecture06, slide 9; Lab6, p. 4]

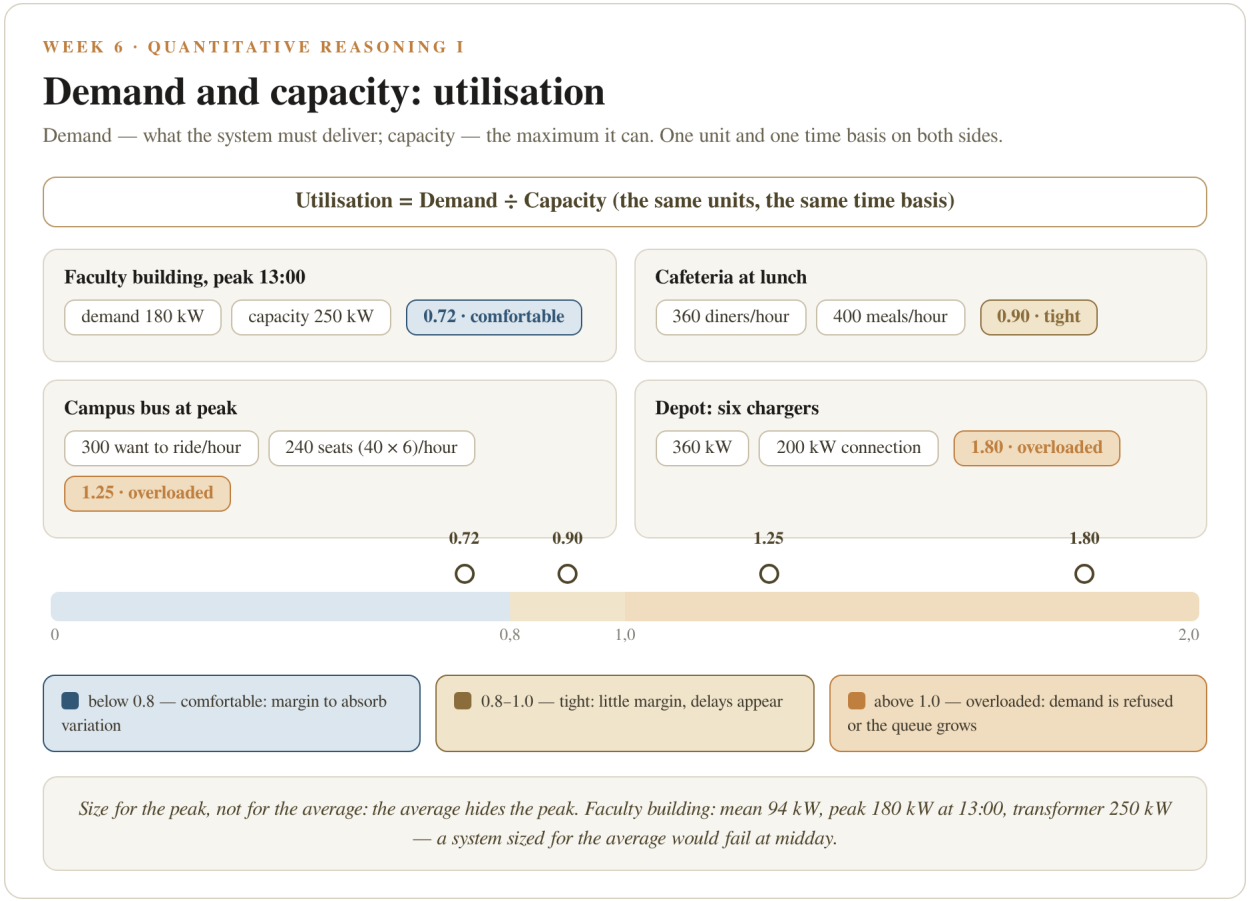


Fig. 3. Demand and capacity (week 6): utilisation = demand ÷ capacity in the four lecture systems; size for the peak, not for the average.

Three ways to close the gap

When demand exceeds capacity, the system has three options. Option 1 — reduce or shift the demand: for example, charging in two batches of three buses. Option 2 — raise the capacity: a larger connection or transformer; this costs money and time. Option 3 — accept the waiting: let the queue grow and decide how much waiting is tolerable. Which option is better is a trade-off; the course comes back to this in week 10.

[Source: Lecture06, slide 10; Lab6, p. 5]

Intensities: an amount per unit of time, and unit traps

An intensity is an amount per unit of time: a speed of 20 km/h; a power of 60 kW ($kW = kWh/h$

- a charger delivers 60 kW); arrivals of 10 students per minute; daily energy of 115 kWh per bus per day. kW is an intensity, kWh is an amount; power $\times$ time = energy. Carry the units in square brackets, as Kosky does: 60 [kW] $\times 2$ [h] $= 120$ [kWh].

Trap	Why it is wrong	How to say it correctly
“The charger consumes 60 kWh”	kWh is energy; the charger’s rating is an intensity	It delivers 60 kW, or 60 kWh every hour
“The bus needs 115”	No unit and no time basis	115 kWh per bus per day
“A 0,2 MW connection is 20 kW”	$MW \rightarrow$ kW is a factor of 1 000	$0,2 MW = 200$ kW
“36 minutes is 0,36 hours”	Minutes are not decimal fractions of an hour	36 min ÷ $60 = 0,6h$

Table 3.2. Unit traps from lecture 6

When an answer looks strange, check the units and the time basis first.

[Source: Lecture06, slides 14, 18]

Stock and flow

Change in stock = flow in − flow out. A stock — a queue, a buffer, a battery — grows when the inflow exceeds the outflow, and is depleted the other way round. Example: driving at 20 km/h with a consumption of 1,2 [kWh/km] $\times 20$ [km/h] $= 24$ [kW] — so 160 kWh lasts about 6,7 h. Buffer filling time: free space ÷ (inflow − outflow).

change in stock = inflow − outflow

[Source: Lecture06, slide 15; Lab6, p. 4]

The week’s hook: six buses, a 200 kW connection

The week’s problem set: six buses, each needing 115 kWh by 06:00, and one 200 kW connection. Is this a problem about energy (kWh) or about power (kW)? First a model example for one bus using Need–Know–How–Solve: Need — the time to restore the daily energy [h]; Know — energy 115 kWh, charger power 60 kW; How — time = energy ÷ power, under the assumptions “constant power, no losses”; Solve — 115 [kWh] ÷ 60 [kW] $\approx 1,9$ [h]; unit check: [kWh] ÷ [kW] = [h]. State the assumption: real charging slows down near full charge and loses some energy, so the true time is longer.

The hook’s answer: the energy is $6 \times 115 = 690$ kWh; the window 22:00–06:$00 = 8h$; the average power is 690 ÷ $8 \approx 86$ kW. All six at once — 360 kW against 200 kW (utilisation 1,8). The solution — shift the demand: two batches of three buses, $3 \times 60 = 180$ kW, utilisation 0,9; the time $2 \times 1,9 \approx 3,8h$ — inside the window. The lecture’s conclusion: “Energy was never the problem. Power was, and staggering the charging solves it”.

[Source: Lecture06, slides 4, 16–17]

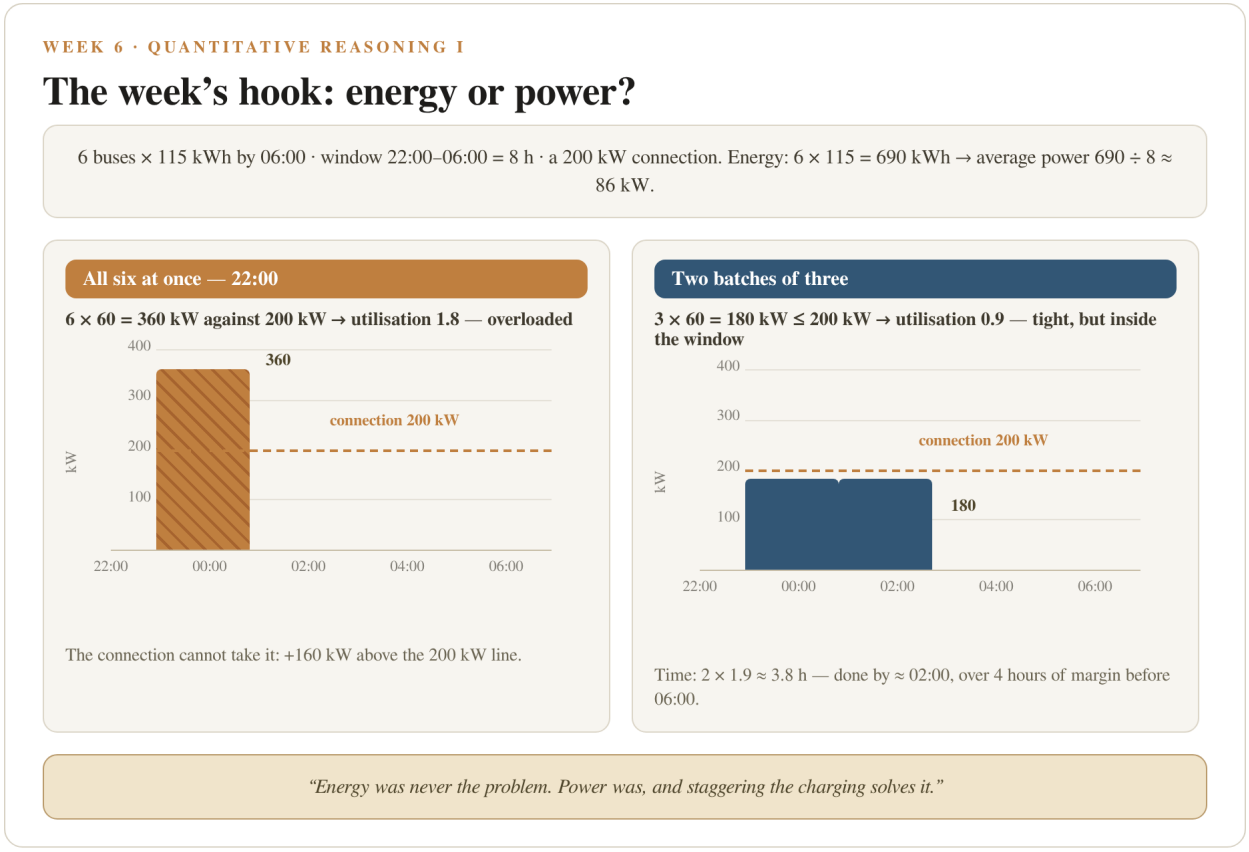


Fig. 4. Six buses, one 200 kW connection (week 6): the 690 kWh of energy was never the problem — power was; two batches of three solve it.

Two rules of flows

Rule 1: Flow = speed × density. Cars passing a point per hour = speed × cars per km: 15 [km/h] ×20 [cars/km] = 300 [cars/h]. Rule 2: number inside = arrival intensity × time inside. People in a queue = arrivals per minute × average time inside: 10 [students/min] ×3 [min] = 30 students in line. Both rules hold under steady-state conditions; the course returns to queues in week 9 (rule 1 — Kosky, ch. 8; rule 2 — a standard result of queueing theory).

Flow = speed × density: 15 [km/h] ×20 [cars/km] = 300 [cars/h]

number inside = arrivals × time: 10 [students/min] ×3 [min] = 30

[Source: Lecture06, slide 19]

Order of magnitude: estimate before you calculate

Estimation helps before the calculation (knowing the expected size: tens, hundreds, thousands?), during it (errors of a power of ten or of units surface immediately) and after it (is the result sensible compared with something known?). An order-of-magnitude estimate is correct to within a factor of ~{}10 and often much closer: Kosky estimated the number of barbershops in a city of 60 000 people at 21, and the phone directory gave 23. The four-step method: (1) identify the quantity and its units; (2) break it into factors you can estimate; (3) round each factor to one digit and multiply; (4) check the units and compare with a known value. Carry the units in square brackets: the units check both the method and the answer.

Thing	Typical size	In other units
LED lamp	10 W	0,01 kW
Electric kettle	2 kW	2 000 W
Phone battery	15 Wh	0,015 kWh
City electric bus battery	200 kWh	0,2 MWh
Faculty building at the peak (our example)	180 kW	0,18 MW

Table 3.3. Everyday anchors for powers and energies ($1kW = 1000W; 1MW = 1000kW; 1kWh =$ 1 000 Wh)

A worked example — solar for one bus: 115 kWh needed per day; 4 kWh per kW of panels per day (a guess, the annual average); the installation size 115 ÷ $4 \approx 29 \rightarrow$ about 30 kW; the roof area 30×5 [m²/kW] $\approx 150m^2$ — "a roof, not a field". The units: [kWh/day] ÷ [kWh/kW-day] = [kW], then [kW] × [m²/kW] = [m²]. Compare with lecture 5: 5 kW gave about 31 kWh on a clear day.

Bound the estimate. Optimistic 5 kWh/kW-day $\rightarrow 115 \div 5 \approx 23$ kW; best guess $4 \rightarrow 29$ kW; cautious $3 \rightarrow 38$ kW. Precise is not the same as correct: report "about 30 kW", not "29,17 kW". From 23 to 38 kW the decision is one — a rooftop installation; when greater precision cannot change the decision, stop refining. Common estimation errors: confusing kW and kWh; forgetting the peak (sizing the connection for the 86 kW average); double counting (seats per bus and per trip counted twice); false precision ("29,17 kW" from rough guesses). Most estimation errors are errors of units or of counting, not of mathematics.

[Source: Lecture06, slides 23–26, 28–29]

3.2 3.2. Savollar of week 6 — lecture and lab questions with model answers

S1. Lecture 6: what are demand and capacity? Why does every number need a time basis?

Answer. Demand is what the system must deliver: the passengers who need a bus at 8:30; the kW consumed by the buses that are switched on. Capacity is the maximum the system can deliver: seats per hour; the kW the connection can provide. The time basis is what the number is per: passengers per hour, kW at 13:00, kWh per day. A number without a time basis is as incomplete as a number without units: 360 kW at 22:00 and 360 kWh per day are different statements.

S2. Name the demand and the capacity in the lecture's four systems.

Answer. The campus bus at the peak: demand — 300 passengers want to ride; capacity — 240 can be carried (40 seats $\times 6$ buses); per hour. The canteen at lunch: 360 diners arrive, 400 meals can be served; per hour. The depot connection: 360 kW if six chargers run, the connection's capacity 200 kW; kW. The faculty building at the peak: 180 kW consumed at 13:00, the transformer rating 250 kW; kW. Always compare like with like: one unit, one time basis.

S3. What is utilisation, what are the thresholds, and what are the values in the four systems?

Answer. Utilisation = demand ÷ capacity. The faculty building: 180 ÷ 250 kW $= 0,72$ — comfortable. The canteen: 360 ÷ 400 per hour $= 0,90$ — tight. The bus: 300 ÷ 240 per hour $= 1,25$ — overload. The depot (all six): 360 ÷ 200 kW $= 1,80$ — overload. Thresholds: below ~{}0,8 — comfortable (margin for variation); ~{}0,8–1,0 — tight

(little margin, delays appear); above 1,0 — overload (demand is turned away or a queue grows).

S4. Why “size for the peak, not for the average”?

Answer. The faculty building’s profile: an average of 94 kW, a peak of 180 kW at 13:00, the transformer 250 kW. A system sized for the average would fail at midday: “Sized for the average, the building would fail at midday”. The average hides the peak. The same rule explains why an average charging power of 86 kW does not guarantee that six 60-kW chargers can run simultaneously: the peak requirement is 360 kW.

S5. In what three ways can the gap between demand and capacity be closed?

Answer. (1) Reduce or shift the demand — for example, charge the buses in two batches of three. (2) Raise the capacity — a larger connection or transformer; this costs money and time. (3) Accept the waiting — let the queue grow and decide how much waiting is tolerable. Which option is better is a trade-off; the course comes back to this in week 10.

S6. Lecture 6, exercise “Demand or Capacity?”: label A–F and compute the utilisation of the pairs.

Answer. A “the depot connection can give 200 kW” — capacity; B “six buses at 22:00 would draw 360 kW” — demand; the pair: 360 ÷ $200 = 1,8$ — overload. C “the stop carries at most 240 per hour” — capacity; D “300 per hour want the 8:30 bus” — demand; 300 ÷ $240 = 1,25$ — overload. E “the faculty transformer is rated 250 kW” — capacity; F “the building’s peak of 180 kW at 13:00” — demand; 180 ÷ $250 = 0,72$ — comfortable. Before dividing, check the unit and the time basis on both sides.

S7. What is the difference between kW and kWh, and how should units be written in calculations?

Answer. kW is an intensity ($kW = kWh/h$), kWh is an amount; power × time = energy. Kosky style: carry the units in square brackets — 60 [kW] $\times 2$ [h] $= 120$ [kWh]; the units check both the method and the answer. Traps: “the charger consumes 60 kWh” — wrong: the rating is an intensity, 60 kW, or 60 kWh every hour; “the bus needs 115” — no unit and no time basis: 115 kWh per bus per day; $0,2MW = 200$ kW (a factor of 1 000); $36\,\text{min} = 0,6h$, not 0,36 h.

S8. What are stock and flow? How many hours does a bus run on 160 kWh at 20 km/h?

Answer. Change in stock = inflow − outflow; a stock (a queue, a buffer, a battery) grows when the inflow exceeds the outflow and is depleted the other way round. At 20 km/h the consumption as power: 1,2 [kWh/km] $\times 20$ [km/h] $= 24$ [kW]; so 160 kWh lasts about 160 ÷ $24 \approx 6,7h$. Buffer filling time: free space ÷ (inflow − outflow).

S9. Lecture 6, model example: how long does it take to charge one bus?

Answer. Need — the time to restore the daily energy [h]. Know — energy 115 kWh; charger power 60 kW. How — time = energy ÷ power; assumptions: constant power, no losses. Solve — 115 [kWh] ÷ 60 [kW] $\approx 1,9$ [h]; unit check: [kWh] ÷ [kW] = [h] — the answer’s unit is the one that was asked for. State the assumption: real charging slows down near full charge and loses some energy, so the true time is longer.

S10. Can six buses charge overnight on a 200 kW connection?

Answer. The energy: $6 \times 115 = 690$ kWh; the window 22:00–06:$00 = 8h$; the average power 690 ÷ $8 \approx 86$ kW — energy is not the problem. The problem is power: all six at once would draw $6 \times 60 = 360$ kW on a 200 kW connection (utilisation 1,8). The solution — shift the demand: two batches of three buses, $3 \times 60 = 180$ kW, utilisation

0,9; the time $2 \times 1,9 \approx 3,8h$ — inside the window. The lecture's conclusion: "Energy was never the problem. Power was, and staggering the charging solves it".

S11. Lecture 6, "Can Ten Buses Charge?": what if the fleet grows to ten buses?

Answer. (1) Total energy: $10 \times 115 = 1150$ kWh. (2) Average power: 1 150 ÷ $8 = 143,75 \approx 144$ kW. (3) At most 200 ÷ $60 = 3,3 \to 3$ chargers can run simultaneously (round down). (4) Batches needed: 10 ÷ $3 = 3,3 \to 4$(round up). (5) Time: $4 \times 1,9 \approx 7,7h$ — under 8 h, the plan fits. But with almost no margin: one slow charger or a late bus would break the plan.

S12. State the two "rules of flows" with examples.

Answer. Rule 1: Flow = speed × density — cars passing a point per hour = speed × cars per km: 15 [km/h] $\times 20$ [cars/km] $= 300$ [cars/h]. Rule 2: number inside = arrival intensity × average time inside: 10 [students/min] $\times 3$ [min] $= 30$ students in line. Both rules hold under steady-state conditions; the course returns to queues in week 9.

S13. What is the order-of-magnitude estimation method, and what does it give?

Answer. Four steps: (1) identify the quantity and its units; (2) break it into factors you can estimate; (3) round each factor to one digit and multiply; (4) check the units and compare with a known value. Estimation helps before the calculation (what order of magnitude to expect), during it (it catches errors of powers of ten and of units) and after it (is the answer sensible). Accuracy — to within a factor of ~{}10 and often better: Kosky estimated 21 barbershops in a city of 60 000, and the directory gave 23. Anchors: a lamp 10 W, a kettle 2 kW, a phone battery 15 Wh, a bus battery 200 kWh, a building peak 180 kW.

S14. Lab 6, the "Semiconductor Wafer Packaging Line" case: where is the bottleneck, what happens to the buffer, and what improvement does the model answer propose?

Answer. Daily capacities of the stages (units/$h \times 12h$): Die Sort 8 160; Die Attach 7 440; Wire Bonding 6 960; Electrical Test 7 200; Final Packaging 7 680. The bottleneck is Wire Bonding (6 960): at an average demand of 6 000 all stages are below $1,0 (WB = 0,86)$; at a peak of $7000, WB = 1,01$ — overload, while the others at 0,86–0,97 are tight. The buffer before WB: change $= 600 - 560 = 40$ units/h; free space $1100 - 300 = 800$; it fills in 800 ÷ $40 = 20h$, and a 12-hour shift adds 480 units (up to 780 of 1 100). Improvement: raise WB capacity to 7 000 ÷ $0,85 \approx 8235$ units/$day \approx 686$ units/h(+18% over 580); the next tightest stage is Electrical Test (7 200), which needs checking.

[Source: Lecture06; Lab6 — lecture and lab questions of week 6]

3.3 3.3. Topshiriqlar of the week 6 lab — with solutions

Lab 6 lasts 80 minutes: a recap of lecture $6 (15$ min$)$ — demand and capacity, utilisation, intensities and flows, peaks, order-of-magnitude checks; instructions $(10$ min$)$; the team work "Quantify the System" $(35$ min, assessed; the slide title says 40 minutes); the Case #1 walkthrough $(20$ min$)$. The agenda announces the task as a "nine-step quantitative task", while the slides and the sheet have ten numbered steps.

[Source: Lab6, pp. 2, 6–8]

Assignment L6-1. Quantify the System — ten steps (assessed)

- Draw the system as a process of 4–6 stages.
- Determine demand and capacity, with units and a time basis.
- Compute the utilisation of the important stages: Utilisation = Demand ÷ Capacity.
- Classify each key stage: comfortable $(< 0,8)$, tight (~{}0,8–1,0),

overload $(> 1,0)$.

- Find the bottleneck and state the maximum practical throughput of the system.
- Compute the material inflow and outflow through the process, including rejects, losses, rework and scrap.
- Compute one queue, buffer or stock change using the formula “change = inflow − outflow”;

find the size from a given time, or the filling time.

- Compare average demand with peak demand; explain whether the system behaves differently at the peak.
- Propose one quantified improvement (reduce/shift the demand, raise the capacity,

accept/manage the waiting); show the new utilisation or capacity.

- An order-of-magnitude check of one answer: the expected scale, the units, whether the result is

plausible.

[Source: Lab6, pp. 7–8]

Model answer L6-1 — Case #1: Semiconductor Wafer Packaging Line

The case conditions. A plant packages power-control chips for industrial electronics. In a regular shift, 7 200 dies arrive per day from wafer production; the line runs 12 h/day. The dies are sorted (Die Sort), attached to a substrate (Die Attach), bonded to the leads (Wire Bonding), tested electrically (Electrical Test) and packed for shipment (Final Packaging). Stage capacities: 680, 620, 580, 600, 640 units/h. Losses: 2% rejects after Die Sort; 3% losses in Die Attach / Wire Bonding; 4% fail Electrical Test (written off). Demand: 6 000 units/day on average; 7 000 units/day in the last 3 days of each month. The buffer before Wire Bonding: currently 300 units, at most 1 100; in the peak hour Die Attach sends 600 units/h and Wire Bonding processes 560 units/h.

Steps 1–2. The process flow and the daily capacity (capacity per day = rate [units/h] $\times 12$ [h/day]).

Stage	Rate [units/h]	Time [h/day]	Capacity [units/day]
Die Sort	680	12	$8160 (= 680 \times 12)$
Die Attach	620	12	7 440
Wire Bonding	580	12	6 960
Electrical Test	600	12	7 200
Final Packaging	640	12	7 680

Table 3.4. Steps 1–2: the stages of the line and their daily capacity (Lab 6, Case #1)

Demand: 6 000 units/day (average) and 7 000 units/day (peak) — both per day, so we compare against the capacity per day.

Steps 3–4. Utilisation and classification of the stages (utilisation = demand [units/day] ÷ capacity [units/day]).

Stage	Capacity [units/day]	Util. at 6 000	Util. at 7 000	Verdict at the peak
Die Sort	8 160	0,74	0,86	tight
Die Attach	7 440	0,81	0,94	tight
Wire Bonding	6 960	0,86	1,01	overload
Electrical Test	7 200	0,83	0,97	tight
Final Packaging	7 680	0,78	0,91	tight

Table 3.5. Steps 3–4: utilisation at average and peak demand; for example, Wire Bonding: 6 000 ÷ $6960 = 0,86 and 7000 \div 6960 = 1,01$

Step 5. The bottleneck. Wire Bonding: 6 960 units/day — the minimum capacity sets the maximum practical throughput of the whole line. At average demand every stage is below 1,0; at the peak four stages are tight, and Wire Bonding is overloaded.

Step 6. The material flow: good chips out of 7 200 dies. $7200 \rightarrow 7056$ after Die Sort $(\times 0,98) \rightarrow 6844$ after attach/bond $(\times 0,97) \rightarrow \approx 6571$ good chips after the test $(\times 0,96)$. Losses per day: 144 rejected in Die Sort, ≈ 212 lost in attach/bond, ≈ 273 written off at the test; about 629 of the 7 200 dies $(\approx 9\%)$ are never shipped.

good output $= 7200 \times 0,98 \times 0,97 \times 0,96 \approx 6571$ [units/day]

Step 7. The buffer before Wire Bonding in the peak hour. Change = inflow − outflow $= 600 - 560 = 40$ units/h. Free space $= 1100 - 300 = 800$ units. Filling time $= 800 \div 40 = 20h$. A 12-hour shift at this rate adds 480 units (up to 780 of 1 100); the buffer fills in 20 h — and then Die Attach is blocked.

Step 8. Average versus peak. Average demand 6 000/day: the system copes — the bottleneck's capacity is 6 960, utilisation 0,86; Wire Bonding is tight, but no stage is overloaded. Peak demand 7 000/day: slightly overloaded — Wire Bonding at 1,01 cannot keep up, the other stages are at 0,86–0,97, and any disturbance builds a queue. Behaviour at the peak differs: the bottleneck saturates, and the buffer fills as in step 7.

Step 9. One quantified improvement — raise the capacity of Wire Bonding. Target: peak utilisation $\leq 0,85$. Required capacity $= 7000 \div 0,85 \approx 8235$ units$/day = 8235 \div$

$12 \approx 686$ units/h. The current 580 units$/h \rightarrow$ an increase of (686 − 580) ÷ $580 \approx 18\%$. The new peak utilisation: 7 000 ÷ $8235 = 0,85$. Electrical Test (7 200 units/day) becomes the next tightest stage — it should be checked.

Step 10. An order-of-magnitude check. The expected scale is thousands per day: about 7 000 dies in, and a few percent losses at each of the three stages $\rightarrow$ an output slightly below 7 000, i.e. thousands, not tens and not millions. The units agree: [units/day] $\times$ [fraction kept, unitless] = [units/day]; the hours: [units/h] $\times$ [h/day] = [units/day] — one time basis on both sides. A rough check: $7000 \times 0,9 \approx 6300$–$6600$; the result ≈ 6571 falls inside the expected range 6 500–7

000. Verdict: 6 500–7 000 units/day is plausible; a result like 65 or 6 500 000 would signal an error of units or of a power of ten. Note: the output (≈ 6571) is above the average demand of 6 000 and below the peak of 7 000 — demand is met on average, falling short only on peak days.

[Source: Lab6, pp. 9–13]

Assignment L6-2. Quantitative Reasoning Worksheet (the week 6 sheet)

The sheet's instruction: "Use the team scenario. Carry the units in [square brackets] and state the time basis of every number. Show the formulas and the working, not only the answers". The sheet's items:

- Process Flow (4–6 stages) — the stages in order, left to right; sketch arrows between them.
- Demand and Capacity — the table "stage / capacity (units/h) / time basis (h/day) / capacity

(units/day)"; average and peak demand separately, with units (one unit and one time basis on both sides, e.g. units/day).

- Utilisation and classification — Utilisation = Demand ÷ Capacity; comfortable $<$ ~{}0,8, tight

~{}0,8–1,0, overload $> 1,0(by$ peak demand).

- Bottleneck and Maximum Practical Throughput — the bottleneck stage, the maximum practical

throughput [units per unit of time] and why it is the bottleneck.

- Material Flow — the table "after stage / fraction lost or rejected / input / lost / output"; the totals:

good output per day and the fraction lost.

- Queue, Buffer or Stock Change — change = inflow − outflow; which queue/ buffer/stock; inflow

[/h], outflow [/h], change [/h]; the current level and the maximum; the size from a given time OR the filling time (with the working).

- Average vs Peak Demand — the bottleneck's utilisation at the average and at the peak; does the system cope;

does the system behave differently at the peak (explain).

- One Quantified Improvement — reduce/shift the demand, raise the capacity, or accept/manage the waiting; what exactly you change; the target utilisation; the calculation; the new capacity or the new utilisation; side effect: which stage is now the tightest.

- Order-of-Magnitude / Reasonableness Check — the answer being checked; the expected scale (tens /

hundreds / thousands $\ldots) and a$ quick estimate; do the units and the time basis work; is the result plausible, and why.

The sheet's model solution — Case #1 "Semiconductor Wafer Packaging Line" (Tables 3.4–3.5 and steps 5–10 of assignment L6-1).

[Source: Lab6WorksheetTemplate.txt]

Assignment L6-3. Lecture 6 exercises (in class, in pairs)

"Demand or Capacity?" — five minutes; watch the units and the time basis. Label each statement D (demand) or C (capacity), then form the pairs and compute the utilisation: A — the depot connection can give 200 kW (capacity) with B — six buses plugged in at 22:00 would draw 360 kW (demand): 360 ÷ $200 = 1,8$, overload. C — the stop can carry at most 240 passengers per hour (capacity) with D — 300 passengers per hour want the 8:30 bus (demand): 300 ÷ $240 = 1,25$, overload. E — the faculty transformer is rated 250 kW (capacity) with F — the faculty building's peak of 180 kW at 13:00 (demand): 180 ÷ $250 = 0,72$, comfortable.

"Can Ten Buses Charge?" — the fleet grows to 10 buses; the same 115 kWh per bus, 60 kW chargers, a 200 kW connection, an 8-hour window; five minutes, units in square brackets. Answers: (1) total energy 1 150 kWh (10×115); (2) average power about 144 kW $(1150 \div 8 = 143,75)$; (3)

3 chargers at a time $(200 \div 60 = 3,3$, round down); (4) 4 batches $(10 \div 3 = 3,3$, round up); (5) time $4 \times 1,9 \approx 7,7h$ — under 8 h, it fits, but almost without margin: one slow charger or a late bus would break the plan.

"Estimate It in Your Team" — choose one question and go through the four steps of the method (identify $\rightarrow$ break into factors $\rightarrow$ estimate and round $\rightarrow$ check the units), writing down the assumptions. A: how many passengers can the campus bus carry per hour? (hint: seats, trips per hour, standing places.) B: how much electricity does the faculty building consume per day? (hint: rooms, lamps, computers, hours of use.) Share in one sentence: "About X, give or take a factor of 3, assuming $\ldots$".

[Source: Lecture06, slides 11–12, 20–21, 27]

Assignment L6-4. Independent Study: Numbers for Your System

- Write down the demand and the capacity of your system, each with a unit and a time basis; compute the

utilisation.

- Do one intensity or flow calculation for your system, carrying the units in square

brackets.

- Do one order-of-magnitude estimate for your system and write out its assumptions.
- Check every result: units, size, and one known value.

Nothing to hand in: this is the team's working material, preparing for next week — limited and incomplete data.
[Source: Lecture06, slide 31]

3.4 3.4. Atamalar — glossary of week 6

Term (RU)	English	One-phrase definition
Demand	Demand	What the system must deliver: 300 passengers per hour, 360 kW
Capacity (throughput)	Capacity	The maximum the system can deliver: 240 passengers per hour, 200 kW
Utilisation	Utilisation	Demand ÷ capacity; $<$ ~{}0,8 comfortable, ~{}0,8–1,0 tight, $> 1,0$ overload
Time basis	Time basis	What the number is per: per hour, at 13:00, per day — without it the number is incomplete
Peak	Peak	The maximum demand (180 kW at 13:00 against an average of 94 kW); size the system for the peak
Rate	Rate	An amount per unit of time: km/h, kW $=$ kWh/h, people$/$ min
Stock and flow	Stock and flow	A stock changes: change $=$ inflow $-$ outflow (a queue, a buffer, a battery)
Buffer	Buffer	A stock between stages; fills when inflow $>$ outflow (40 units$/h \rightarrow$ full in 20 h)
Bottleneck	Bottleneck	The stage with the smallest capacity (Wire Bonding 6 960 units/day); sets the system's maximum
Order of magnitude	Order of magnitude	An estimate correct to a factor of ~{}10: 21 barbershops against 23 in the directory

Term (RU)	English	One-phrase definition
Everyday anchor	Everyday anchor	Everyday scales for checking: a lamp 10 W, a kettle 2 kW, a bus battery 200 kWh
Bounding	Bounding	A low / typical / high estimate ($\frac{23}{29}/38kW$); stop when the decision no longer changes
False precision	False precision	"29,17 kW" from rough guesses; keep one–two significant figures
Queue	Queue	The consequence of utilisation above 1: demand is turned away or waits; in detail in week 9

[Source: Lecture06; Lab6]

4 4. Week 7 — Quantitative Reasoning II

4.1 4.1. Key statements (“all the qoidalar” of week 7)

The week 7 lecture teaches you to reason when data are scarce, and to read the same numbers in more than one way. A recap of week 6 “in one breath”: demand and capacity — what is asked against what can be delivered, with units and a time basis; intensities and flows — an amount per unit of time, power × time = energy, change = inflow − outflow; estimation — break into factors, round to one digit, check the units; bounding — low, typical, high, and stop when the decision no longer changes.

Five ways data can be incomplete

The week’s hook: three test trips — 1,1; 1,3 and 1,6 kWh/km. Which number should be used for the range promise? Before choosing a number, name the data problem — there are five:

Problem	What it means	Example with the bus
Too little	Only a few points	Three test trips
Missing	No data for the case we care about	No winter test
Noisy	Values scatter around the truth	1,1–1,6 kWh/km on the same route
Outdated / unrepresentative	Measured under other conditions	Tests with an empty bus
Wrong definition or unit	The quantity is not what we think	The battery in “%” or “200” without a unit

Table 4.1. Five ways data can be incomplete (Lecture07, slide 6)

Name the problem first. The way to fix it depends on which of the five it is.

[Source: Lecture07, slides 4, 6; Lab7, p. 3]

Where numbers come from, and how far to trust them

Source	Example	How far to trust
Measure	Time a bus trip, keep a charging log	High, if the method is stated

Source	Example	How far to trust
Look up	A datasheet, a standard	Average: conditions may differ
Ask	The driver, the operations manager	Average: check it against something
Estimate	Break into factors (lecture 6)	Low–medium: state the assumptions
Analogy	A similar building or route	Low: say how similar

Table 4.2. Sources of numbers and the level of trust (Lecture07, slide 7)

Write the source and your confidence next to every number.

[Source: Lecture07, slide 7; Lab7, p. 3]

Ranges instead of single numbers

Three summer trips give a consumption of 1,1 / 1,3 / 1,6 kWh/km and a range of 160 ÷ $e = \frac{145}{123}/100$ km. All three cases cover the daily requirement of 96 km, but the high-consumption case — only with a 4 km margin. Report: “about 100 to 145 km”. A range tells the reader how far to trust the answer.

$R = 160$ [kWh] ÷ e: 1,1 / 1,3 / 1,6 [kWh/km] → $\frac{145}{123}/100$ [km]

[Source: Lecture07, slide 8]

Small samples: state n

Three trips: the mean $= (1,1 + 1,3 + 1,6) \div 3 = 1,33$ kWh/km — one odd trip moves it a lot; the median $= 1,3$ kWh/km — the middle element in order, which one odd trip barely moves; the sample size $n = 3$ — write n next to every number you report. With three points, give a range and n; do not claim more than three trips can show.

mean $= (1,1 + 1,3 + 1,6) \div 3 = 1,33$ [kWh/km]; median $= 1,3$ [kWh/km]; $n = 3$

[Source: Lecture07, slide 9; Lab7, p. 4]

Triangulation: estimate in three ways

Triangulation is estimating one quantity by several independent routes and comparing. The consumption of the faculty building, kWh/day: route 1 — floor area × typical consumption: 8 000 [m²] $\times 0,35$ [kWh/m²·day] ≈ 2800; route 2 — the load profile from lecture 6: 94 [kW] $\times 24$ [h] ≈ 2260; route 3 — the monthly bill: 70 000 [kWh/month] ÷ $30 \approx 2330$. Three independent routes agree to within about 25%. Agreement builds trust; disagreement points to where to look.

$\approx 2800 = 8000 \times 0,35$; $\approx 2260 = 94 \times 24$; $\approx 2330 = 70000$ ÷ 30 [kWh/day]

[Source: Lecture07, slide 10; Lab7, p. 4]

One dataset — different readings

The distribution of waiting times at a stop (share of passengers by waiting interval 0–$2 \ldots 14$–16 minutes: 14, 26, 22, 12, 7, 4, 8, 7%). The dispatcher: “The average wait is 6 minutes. Good.” The passenger: “Every seventh of us waits more than 12 minutes” ($\frac{1}{7} \approx 15\% = 7\% + 8\% of the 12$–$14 and 14$–$16$ intervals). The administrator: “Cutting the long waits requires more buses. What does that cost?” One set of numbers, three readings — and all three are true.

[Source: Lecture07, slide 14]

The metric is chosen by the decision

Question	Metric	Value here
How long does the typical passenger wait?	Median	about 5 min
How much waiting in total?	Mean	about 6 min × passengers
How bad is it for the unlucky?	90th percentile	about 13 min
What is the worst case?	Maximum	about 16 min

Table 4.3. The decision chooses the metric (Lecture07, slide 15)

The decision chooses the metric, not the other way round. Lecture 6 had the same pair: kWh answers “how much?”, kW — “how fast?”.

[Source: Lecture07, slide 15; Lab7, p. 5]

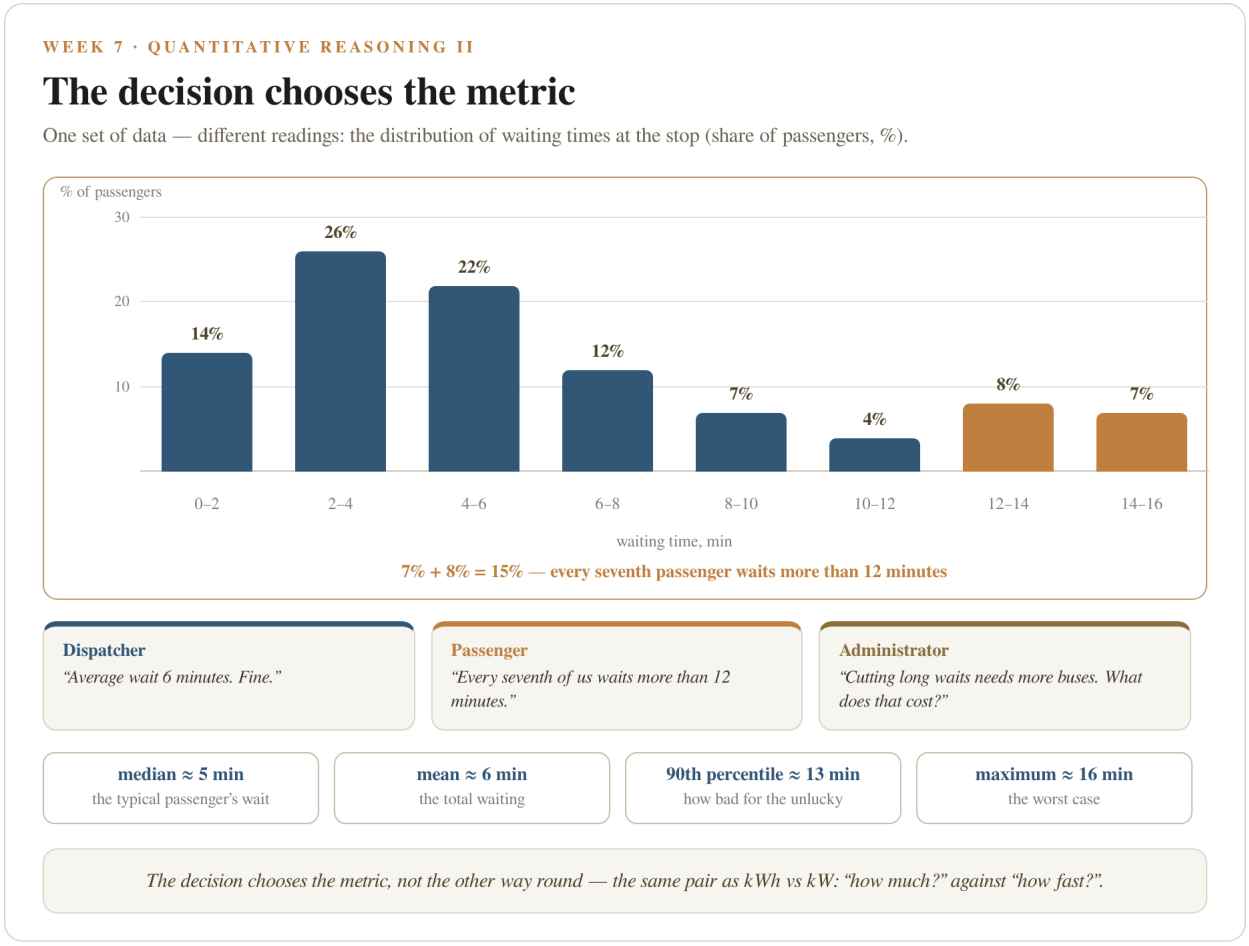


Fig. 6. The decision chooses the metric (week 7): one waiting-time distribution — three readings and four metrics.

The baseline changes the story

One fact: the LED retrofit saves 12 kW. Two true percentages: 60% of corridor lighting (12 ÷ 20 kW) and 7% of the building's peak (12 ÷ 180 kW). Different baselines and boundaries give different but true stories. Always ask: a percentage of what?

Misleading comparisons need no false number — each needs a question. Different denominators: "Route A has fewer complaints" (it has far fewer passengers) — ask "per what?". Cherry-picked days: "the best week of the semester" — "which days were dropped?". A truncated axis: a bar chart starting at 80 turns a 5% gap into a huge one — "where does the axis start?". Four questions for any interpretation: (1) for which decision is this number? (2) which metric fits the decision — the mean, the peak, a percentile, the worst case? (3) which baseline, boundary and time basis are used? (4) which assumption, if it turned out wrong, would flip the conclusion? Ask them before accepting a result, and before presenting it.

[Source: Lecture07, slides 16–18]

Honest reporting and the "winter day" summary example

An honest engineering report consists of four elements: RESULT — the answer, a simple sentence the reader can use; RANGE — how far to trust it: low to high, not a single value; TOP ASSUMPTION — the assumption the result rests on most; WHAT

WOULD CHANGE IT — what to find out next: the data that would move the conclusion.

The lecture's summary example: can one bus get through a winter day of 96 km on a single charge? Known: usable energy 160 kWh, daily requirement 96 km, summer tests 1,1–1,6 kWh/$km(n = 3)$. Missing: a winter test — cabin heating raises consumption by an unknown amount, and it cannot be measured yet. The gap is filled with a range: winter consumption 1,5–2,0, typically 1,8 kWh/km

- an estimate by analogy "summer + heating", low confidence. The calculation: $160 \div 1, \frac{5}{1}, \frac{8}{2}, 0 = \frac{107}{89}/80$ km — only the optimistic case passes. The report: result — "probably not: 89 km against 96 km"; range 80–107 km; top assumption — 1,8 kWh/km with low confidence; what would change it — one real winter trip or daytime top-up charging. The decision: plan daytime top-ups and collect winter data before making commitments. Low confidence goes on the page: hiding it does not remove it.

winter range $= 160$ [kWh] ÷ 1,5 / 1,8 / 2,0 [kWh/km] $= \frac{107}{89}/80$ [km] against 96 [km]

[Source: Lecture07, slides 21–26; Lab7, p. 6]

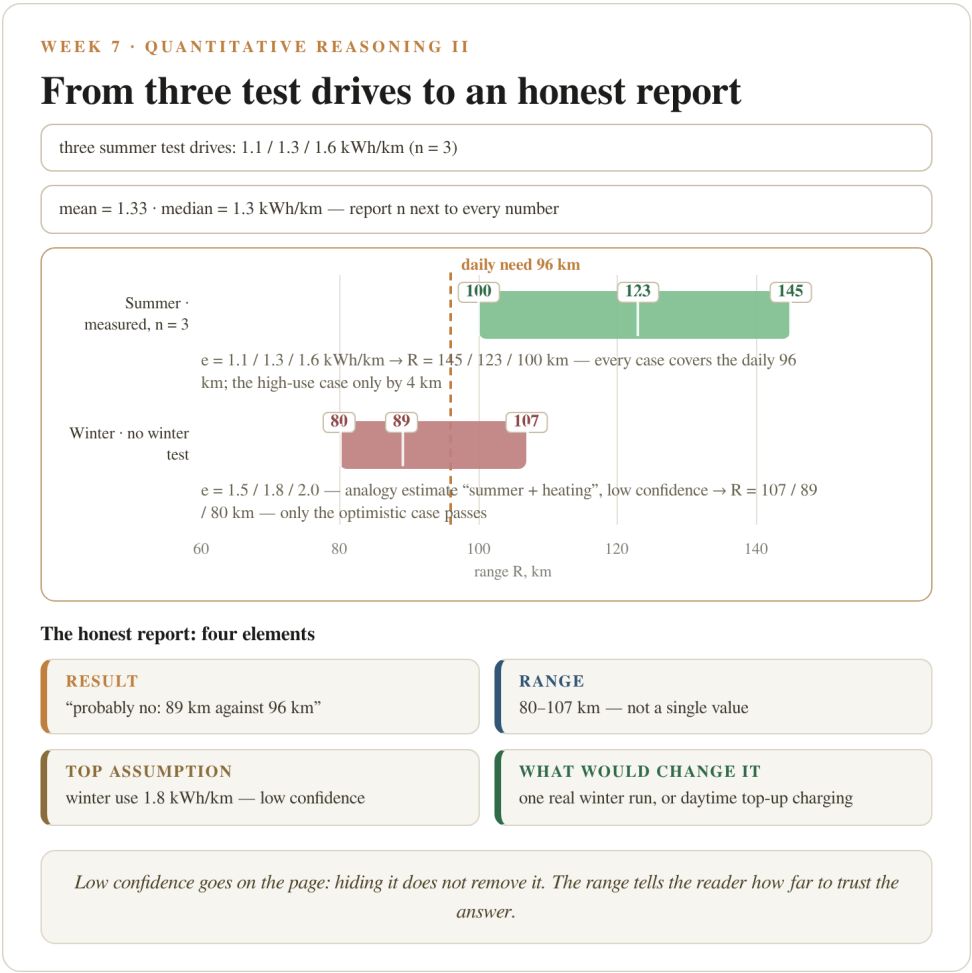


Fig. 5. Ranges and the honest report (week 7): from three summer test drives and the winter gap to the RESULT · RANGE · TOP ASSUMPTION · WHAT WOULD CHANGE IT report.

4.2 4.2. Savollar of week 7 — lecture and lab questions with model answers

S1. Name the five ways data can be incomplete, with bus examples.

Answer. (1) Too little — a few points: three test trips. (2) Missing — no data for the case we need: no winter test. (3) Noisy — values scatter around the truth: 1,1–1,6 kWh/km on the same route. (4) Outdated or unrepresentative — measured under other conditions: tests with an empty bus. (5) Wrong definition or unit — the quantity is not what we think: the battery in "%" or "200" without a unit. Name the problem first: the fix depends on which of the five it is.

S2. Where do numbers come from, and how far should they be trusted?

Answer. Measure (time a trip, keep a charging log) — high confidence, if the method is stated. Look up (a datasheet, a standard) — average: conditions may differ. Ask (the driver, the operations manager) — average: check it against something. Estimate (break into factors, lecture 6) — low–medium: state the assumptions. Analogy (a similar building or route) — low: say how similar. Write the source and the confidence next to every number.

S3. Why a range rather than a single number? What should be reported for the three summer trips?

Answer. A range tells the reader how far to trust the answer. Three trips of 1,1 / 1,3 / 1,6 kWh/km give a range of 160 ÷ $e = \frac{145}{123}/100$ km — all three cases cover the daily requirement of 96 km, but the high-consumption case only with a 4 km margin. Report: "about 100 to 145 km". A single number (for example 123 km) would hide how thin the margin is in the worst case.

S4. Mean or median for a small sample? What should be written next to the number?

Answer. Three trips: the mean $(1,1 + 1,3 + 1,6) \div 3 = 1,33$ kWh/km; the median — the middle element in order — 1,3 kWh/km. One odd trip moves the mean a lot, and the median almost not at all. With $n = 3$, give a range and n; do not claim more than three trips can show. The rule: write n next to every reported number.

S5. What is triangulation? Give the faculty building example.

Answer. Triangulation is estimating one quantity by several independent routes and comparing. The faculty building, kWh/day: route 1 — floor area × typical consumption: 8 000 m² $\times 0,35$ kWh/m²$\cdot day \approx 2800$; route 2 — the load profile of lecture 6: 94 kW $\times 24h \approx 2260$; route 3 — the monthly bill: 70 000 ÷ $30 \approx 2330$. Three independent routes agree to within about 25%: agreement builds trust, disagreement points to where to look.

S6. Lecture 7, exercise "Name the Problem": name the problem type A–F and the way to get a better number.

Answer. A "We tested the bus in one trip" — too little: repeat the trip on different days and loads. B "No data on winter heating of the dormitory" — missing: estimate by analogy; schedule a winter measurement. C "Five charger readings: 58, 61, 54, 63, 60 kW" — noisy: the mean and a range, about 59 kW (54–63). D "The generation figure is from a desert site" — unrepresentative: find data from a similar climate. E "The datasheet says '200' without a unit" — wrong definition/unit: 200 kWh is energy, 200 kW is power. F "The load profile is from before the new wing" — out of date: re-measure or scale the profile.

S7. One dataset — three readings: how do the dispatcher, the passenger and the administrator read the waiting distribution?

Answer. The dispatcher: “The average wait is 6 minutes. Good.” The passenger: “Every seventh of us waits more than 12 minutes” — $\frac{1}{7} \approx 15\%$, the $12–14\,\text{min}\,bar(7\%)andthe14–16\,\text{min}\,bar(8\%)$. The administrator:

“Cutting the long waits requires more buses. What does that cost?” One set of numbers, three readings — and all three are true. Hence the rule: before accepting a number, ask for which decision it is being used.

S8. How is the metric chosen by the deciding question? Give the waiting table.

Answer. “How long does the typical passenger wait?” — the median, about $5\,\text{min}$. “How much waiting in total?” — the mean, about $6\,\text{min} \times$ passengers. “How bad is it for the unlucky?” — the 90th percentile, about $13\,\text{min}$. “What is the worst case?” — the maximum, about $16\,\text{min}$. The decision chooses the metric, not the other way round. Lecture 6 had the same pair: kWh answers “how much?”, kW — “how fast?”.

S9. “A percentage of what?” — explain it with the LED retrofit savings.

Answer. One fact: the LED retrofit saves 12 kW. Two true percentages: 60% of corridor lighting (12 ÷ 20 kW) and 7% of the building’s peak (12 ÷ 180 kW). Different baselines and boundaries give different but true stories — always ask, a percentage of what. In the lab’s LED case the same saving of 38 kW gives 33% (against the 117 kW of old connected lighting — the same boundary, the same percentage), 51% (against the operations estimate of 75 kW — too high) and 191% (against the old report of 20 kW — impossible: the boundary is wrong).

S10. Which three techniques mislead without a single false number, and which questions counter them?

Answer. Different denominators: “Route A has fewer complaints” — while it has far fewer passengers; ask “per what?”. Cherry-picked days: “the best week of the semester” — “which days were dropped?”. A truncated axis: a bar chart starting at 80 makes a 5% gap huge — “where does the axis start?”. None of these techniques needs a false number; each needs a question.

S11. What four questions does the lecture ask of any interpretation of a number?

Answer. (1) For which decision is this number? (2) Which metric fits that decision: the mean, the peak, a percentile, the worst case? (3) Which baseline, boundary and time basis are used? (4) Which assumption, if it turned out wrong, would flip the conclusion? Ask these questions before accepting a result — and before presenting it.

S12. What does an honest engineering report consist of?

Answer. Four elements: RESULT — the answer, a simple sentence the reader can use; RANGE — how far to trust it: low to high, not a single value; TOP ASSUMPTION — the assumption the result rests on most; WHAT WOULD CHANGE IT — the data that would move the conclusion. A good report gives the answer, the degree of trust, and what would change it. Low confidence goes on the page: hiding it does not remove it.

S13. The lecture’s summary example: can a bus get through a winter day of 96 km on a single charge?

Answer. Known: usable energy 160 kWh, requirement 96 km, summer tests 1,1–1,6 kWh/km. Missing: a winter test — cabin heating raises consumption by an unknown amount. The gap is filled with a range: winter consumption 1,5–2,0, typically 1,8 kWh/km (the analogy “summer + heating”, low confidence). The calculation: 160 ÷

$1, \frac{5}{1}, \frac{8}{2}, 0 = \frac{107}{89}/80$ km — only the optimistic case passes. The report: “probably not: 89 km against 96 km; range 80–107 km; top assumption — 1,8 kWh/km, low confidence; what would change it — a real winter trip or daytime top-up charging”. The decision: plan the top-ups and collect winter data before making commitments.

S14. Lab 7, the LED case: what is the range of annual savings, and what would flip the “replace everything now” decision?

Answer. The range: 124–216 MWh/year, typically $167 (n = 5$ lamps from one batch); the cost 87–216 million UZS/year; the most important uncertainty is the hours of use. The energy claim is robust: it would flip only if the new lamp really draws close to 18 W (the tests: 11,7–12,6 W). “Replace now” depends on service life and price: the break-even premium ≈ the annual saving per lamp × years of service $(25,8 kWh \times 850 UZS \approx 21900 UZS/$year): $40000h \approx 200000 UZS$; $20000h \approx 100000$; $10000h \approx 50000$. The recommendation — a pilot in one building before replacing all 6 500 lamps.

[Source: Lecture07; Lab7 — lecture and lab questions of week 7]

4.3 4.3. Topshiriqlar of the week 7 lab — with solutions

Lab 7 lasts 80 minutes: a recap of lecture $7 (15$ min) — incomplete data, ranges, triangulation, metrics and baselines, honest reporting; instructions $(10$ min) — the nine-part A–I template for deciding with incomplete data; the team work “Decide With Incomplete Data” $(35$ min, assessed) — diagnose, range, interpret, triangulate, find the break point, report honestly; the Case #1 walkthrough $(20$ min) — LED Lamp Reliability and Energy-Saving Decision.

[Source: Lab7, p. 2]

Assignment L7-1. Decide With Incomplete Data — parts A–I (assessed)

- A. Diagnose the data problems — at least 3 problems; name each one: too little, missing,

noisy, out of date / unrepresentative, wrong definition / unit; why it is a problem, and how to get a better number.

- B. Source and confidence — choose 4–5 important numbers; for each: the value or range,

the source (measure, look up, ask, estimate, analogy), the confidence, and a short reason.

- C. Build a low / typical / high range — choose one significant uncertain

variable; recompute the main result for each value; write: “the output is roughly __ to __, typically __”.

- D. One dataset — two interpretations — a favourable and a critical reading; which numbers

support each; what additional data would settle the dispute.

- E. Choose the right metric — for 3 deciding questions choose average, median, 90th

percentile, maximum, peak or rate per unit / user; say why — and why another metric could mislead.

- F. Check the baseline / denominator — express one result against two different bases;

show both calculations; explain how both percentages can be true yet tell different stories.

- G. Triangulate one key quantity — estimate it by at least 2 independent methods;

do the estimates agree closely enough to support the decision?

- H. What would flip the decision? — find the most important uncertain assumption and the threshold at

which the decision changes; show the calculation or the reasoning.

- I. The final honest report — result, range, top assumption, what would change the conclusion —

in one sentence; then the 60-second team report.

[Source: Lab7, pp. 7–8; Lab7EngineeringDecisionsIncompleteDataTemplate.txt]

Model answer L7-1 — Case #1: LED Lamp Reliability and Energy-Saving Decision

The case task. Decide how confidently the university can estimate the annual energy saving, and whether replacing all the lamps right now is justified.

A. Diagnosing the data problems — five problems; the type of each determines the fix.

Quantity / statement	Problem type	Why it is a problem	A better number
Only 5 lamps tested	too little; unrepresentative	$n = 5$ from one batch; only 2 lamps have run more than 2 000 h	Test several batches and more lamps
40 000 h service life	missing	The supplier's datasheet; no university evidence yet	An accelerated life test; field failure data
6 000 lamps	out of date	The count predates the opening of the new block; now 6 300–6 700	Recount from the asset register
20 kW against 75 kW	wrong definition / boundary	Different baselines of "corridor lighting"	Ask what is included in each; check bottom-up $(6500 \times 18W \approx 117kW)$
10–14 h/day; 700–1 000 UZS/kWh	noisy (a wide spread)	Estimates, not logs; the tariff depends on the contract period	Log the hours; use the contract tariff

Table 4.4. Part A: diagnosing the data of the LED case (Lab 7, Case #1)

B. Source and confidence. The numbers by strength: the power ones are the strongest, the hours and the service life are the weakest; this decides what to check next.

Quantity	Value / range	Source	Confidence	Short reason
Power of the new lamp	11,7–12,6 W; mean $12,1 W (SD 0,35, n = 5)$	measurement	average	The method is stated, but one batch and $n = 5$
Power of the old lamp	18 W	rating (look up)	average	A rating, not a measurement; check several old lamps
Number of lamps	6 300–6 700 (was 6 000)	a technician's estimate	medium-low	An old count; a new block has opened
Hours of use	10–14 h/day	estimate (ask)	low-medium	No logs; $\pm 17\%$ around 12 h
Service life	$40000 h (\approx 9$ years at 12 h/day)	the supplier's datasheet	low-medium	Not verified locally; 2 lamps beyond 2 000 h

Table 4.5. Part B: source and confidence for the five key numbers

C. A low / typical / high range. Energy saving = saving per lamp $\times$ lamps $\times$ hours $\times 365 \div 1\,000$.

Case	Saving per lamp [W]	Lamps [units]	Hours [h/day]	Tariff [UZS/kWh]	Energy saving [MWh/year]	Cost [million UZS/year]
Low	5,4 (18 − 12,6)	6 300	10	700	124	87
Typical	5,88 (18 − 12,12)	6 500	12	850	167	142
High	6,3 (18 − 11,7)	6 700	14	1 000	216	216

Table 4.6. Part C: the annual saving as a range (Lab 7, Case #1)

The typical case: 5,88 [W] $\times 6500 \times 12$ [h/day] $\times 365 \div 1000 \approx 167000$ [kWh/year]. The range statement: about 124 to 216 MWh per year, typically $167 (n = 5$ lamps, one batch); in cost — about 87 to 216 million UZS per year. Sensitivity, one variable at a time (the others typical): hours of use 139–195 MWh/year $(10 \to 14 h/day)$ — the most important uncertainty; the per-lamp power saving 154–179 MWh/year $(5,4 \to 6,3 W)$; the number of lamps 162–173 MWh/year $(6300 \to 6700)$.

D. One dataset — two interpretations. The favourable reading (the procurement manager): "Replace everything" — all five new lamps measure about 12 W against 18 W: a saving of about a third per lamp (30–35%), very likely real. The critical reading (the operations engineer): "We do not know they will last that long" — the service life is unverified, the number of lamps and the hours are uncertain, and an energy saving alone does not prove a lower life-cycle cost. The numbers supporting both: the measured power and the saving — 12 W against $18 W (\approx 6 W per$ lamp) are true in both readings; the dispute is about service life, hours and price, not about watts. The data that would settle it: a multi-batch field pilot with failure logs, logged hours of use, the current number of lamps, the purchase and installation price.

E. Choosing the metric for the decision.

Deciding question	Best metric	Why; how another metric would mislead
How much energy will be saved?	Annual kWh saved (and the range)	It uses the real number of lamps and hours; a percentage per lamp hides how many lamps there are
What does the typical new lamp draw?	The median (with the mean, range and n)	With $n = 5$ the mean and the median coincide (12,1 W); state n and show the range
Will the lamps last long enough?	Failure rate / hours of operation	Average power says nothing about reliability; one datasheet figure is not evidence
Does the replacement pay off?	Payback or cost per year of service	The energy saving ignores the purchase price and early failures

Table 4.7. Part E: the decision chooses the metric (Lab 7, Case #1)

F. Checking the baseline. The saving per lamp of 5,88 W against the old lamp's 18 W: 5,88 ÷ $18 = 33$% — the "third" of purchases is true per lamp (30–35%). The full saving ≈ 38 kW ($6500 \times 5,88W$, all included) against the old connected lighting of $6500 \times 18W \approx 117$ kW: 38,2 ÷ $117 = 33$% — the same boundary, the same percentage. The same saving of 38 kW against the operations estimate of 75 kW: 38,2 ÷ $75 = 51$% — too high: 75 kW cannot be all the lamps on at once. Against the old report's 20 kW: 38,2 ÷ $20 = 191$% — impossible (above 100%): the baseline boundary is wrong. The rules: a result above 100% is a hint that the baseline boundary is wrong; use one boundary before and after the retrofit and state it: corridor lamps only, all lighting, or the whole building.

G. Triangulating the annual saving. Route 1, bottom-up (lamps): measured power × counted lamps × hours — $6500 \times 5,88W \times 12h \times 365 \div 1000 \approx 167$ MWh/year, confidence medium-low. Route 2, top-down (75 kW): the operations estimate $\times a$ one-third saving — 75 kW $\times 0,327 \times 12h \times 365 \approx 107$ MWh/year, low-medium. Route 3, top-down (20 kW): the old energy report × one third — 20 kW $\times 0,327 \times 12h \times 365 \approx 29$ MWh/year, low. An agreement check: routes 1 and 2 agree to within a factor of 1,6 (one order of magnitude) — the saving is real and large, but its exact size is not firm; route 3 is six times lower — that would mean only about 1 100 lamps are on, so the 20 kW figure probably covers only a subset. Next step: find out how many lamps are on at night and for how many hours.

H. What would flip the decision. The energy claim is robust: it would flip only if the new lamp really draws close to 18 W; the tests show 11,7–12,6 W — the energy saving is beyond doubt. The "replace now" decision depends on service life and price: the break-even premium per lamp ≈ the annual saving per lamp × years of real service. The typical saving: 25,8 kWh $\times 850 UZS \approx 21900 UZS$ per lamp per year (range 13 800–32 200). If the lamp really lasts 40 000 h (the supplier's claim) $\approx 9,1$ years $\rightarrow$ the break-even premium $\approx 200000 UZS$; 20 000 h (half the claim) $\approx 4,6$ years $\rightarrow \approx 100000 UZS$; 10 000 h (a quarter) $\approx 2,3$ years $\rightarrow \approx 50000 UZS$. The price of the lamp and installation is not given, so the threshold is a formula, not a single number. The best next measurement: a multi-batch field pilot.

I. The final honest report. The result: the saving is real, the payback is unproven — about a third less energy per lamp, ≈ 167 MWh/year across the campus. The range: 124–216 MWh/year; the top-down check gives ≈ 107; the saved cost 87–216 million UZS/year. The top assumption: the hours and the real service life — 12 h/day and 40 000 h are both unverified. What would change it: a pilot, a recount, the price — logged hours, the current number of lamps, failure data from several batches, the

price of the lamp. The one-sentence report: “The new lamps should save about 170 MWh a year (range about 125 to 215, from 5 lamps of one batch), but because the 40,000 h life and the price are unverified we recommend a one-building pilot before replacing all 6,500 lamps”.
[Source: Lab7, pp. 10–18]

Assignment L7-2. Group Work Template: Engineering Decisions with Incomplete Data (the week 7 sheet)

The template’s goal: “Use the scenario data carefully. Do not assume that every number is equally reliable. Diagnose the data, use ranges, compare interpretations, and report what you can conclude — and what you cannot”. The structure repeats parts A–I of assignment L7-1; the sheet sets the formats to fill in:

- Diagnosis: the table “quantity / statement — problem type — why it is a problem — how to

improve it”.

- Source and confidence: the table “quantity — value or range — source — confidence —

short reason”.

- Range: the table “case (low / typical / high) — input value — unit —

calculated output — decision / interpretation”; the phrase: “The output is approximately __ to __, with a typical value of __”.

- Interpretations: favourable, critical; which numbers support each; what additional data would settle the dispute.

- Metric: the table “deciding question — chosen metric — why — why another metric

could mislead”.

- Baseline: the table “result — baseline — calculation — percentage — interpretation”.

- Triangulation: the table “route — method / source — working — result with unit —

confidence” + an agreement check.

- Break point: “the most important uncertain variable / assumption — the current value or

range — the threshold that would change the decision”.

- Final report: RESULT / RANGE / TOP ASSUMPTION / WHAT WOULD CHANGE *IT+*

a one-sentence report.

- The 60-second team report: “Our current decision is …; the result is in the range from …to…;

the weakest number in our analysis is …; the one piece of data we would collect next,

- …”.

The lecture 7 principle printed on the sheet: a good engineering message names the result, the range, the top assumption, and what would change it.
[Source: Lab7EngineeringDecisionsIncompleteDataTemplate.txt]

Assignment L7-3. Lecture 7 exercises (in class and interactive)

“Name the Problem” (on your own, on paper, five minutes) — name the problem type for each statement and one way to get a better number. Answers: A “We tested the bus in one trip” — too little: repeat the trip on different days and with different loads. B “No data on winter heating of the dormitory” — missing: estimate by analogy with other buildings; schedule a winter measurement. C “Five readings of one charger: 58, 61, 54, 63, 60 kW” — noisy: report the mean and the range — about 59 kW, from 54 to 63. D “The solar generation figure is from a desert site” — unrepresentative: find generation data for a similar climate. E “The datasheet says ‘200’ without a unit” — wrong definition / unit: check the label — 200 kWh is energy, 200 kW is power. F “The load profile is from before the new wing opened” — out of date: re-measure, or scale the profile to the new wing. The problem type suggests which fix to try.

“Read It Twice” (on your own, six minutes). Route data: the average wait 6 min; 15% of passengers wait more than 12 min; 82% of trips on time; at the peak 300 passengers per hour want the bus, capacity 240. (1) One favourable sentence, as the dispatcher would say it: “82% of trips are on time, and the average wait is only 6 minutes” (what would settle it: waits by hour of day). (2) One critical one, as the passenger would say it: “At the peak 300 of us want the bus, and only 240 can ride. Every seventh person waits more than 12 minutes” (what would settle it: the share of passengers left behind at the peak). Both sentences are true — additional data decide which problem matters more.

“The Winter Range Explorer” (interactive, about 3 minutes, via the link in the notes): (1) select “Summer tests” — which trips cover the 96 km requirement? By the lecture slides, all three summer cases: 1,1 / 1,3 / 1,6 kWh/km give 145 / 123 / 100 km — all cover it. (2) select “Winter and aged battery” — by how many km does it fall short in the best case? (3) from “Winter guess”, raise the usable share until the verdict changes — what share is needed? The answers to questions 2–3 are explored on the interactive page (the values are illustrative); for comparison: in lecture 5, winter with an old battery gave 71 km — 25 km below the requirement.

“Your Turn: Rooftop Solar for the Building” (on your own, six minutes, units in square brackets). Can a 30-kilowatt rooftop installation cover the building’s daily electricity? Given: the building averages 94 kW (lecture 6); the installation yields 3–5 kWh per kW per day. Answers: (1) the installation’s energy per day: 30 [kW] $\times 3$–5 [kWh/kW·day] $= 90$–150 kWh (typically 120); (2) the building’s energy per day: 94 [kW] $\times 24$ [h] ≈ 2260 kWh; (3) the covered share: 90 ÷ 2 260 and 150 ÷ $2260 \approx 4$–7% (typically 5%); (4) the report: “The installation covers about 4–7% of the building’s daily electricity, assuming 3–5 kWh per kW per day. Measured generation would

change this". Both statements are true: a few percent of the building — but about one bus per day.

[Source: Lecture07, slides 11–12, 19–20, 27–29]

Assignment L7-4. Independent Study: Incomplete Data in Your System

- Choose one quantity of your system where the data are incomplete, and name the problem type.
- Write down a low, a typical and a high value, with the source and your confidence.
- Show how the range changes one result.
- Write one report sentence: result, range, top assumption, what would change

the conclusion.

Nothing to hand in: this is working material for your system, ready for the second half of the course.

[Source: Lecture07, slide 31]

4.4 4.4. Atamalar — glossary of week 7

Term (RU)	English	One-phrase definition
Incomplete data	Incomplete data	The needed numbers are absent or cannot be fully trusted — five typical problems
Too little data	Too little	Only a few points: three test trips
Missing data	Missing	No data for the case we care about: no winter test
Noisy data	Noisy	Values scatter around the truth: 1,1–1,6 kWh/km on the same route
Outdated / unrepresentative data	Out of date / unrepresentative	Measured under other conditions or earlier: empty-bus tests, a profile from before the new wing
Wrong definition or unit	Wrong definition or unit	The quantity is not what we think: “200” without a unit, the battery in “%”
Triangulation	Triangulation	Estimate one quantity by at least two independent routes and compare $(\approx 2\frac{800}{2}\frac{260}{2}330kWh/day)$
Range	Range	Low–typical–high $(\frac{107}{89}/80km)$; says how far to trust the answer
Median	Median	The middle element in order; robust to one odd value (1,3 kWh/km)
90th percentile	90th percentile	The value below which 90% of observations fall: about 13 minutes of waiting
Baseline (denominator)	Baseline / denominator	“A percentage of what?”: 12 kW $= 60$% of corridor lighting, but 7% of the building’s peak
Top assumption	Top assumption	The assumption the result rests on most: 1,8 kWh/km in winter
Flip point	Flip point	The threshold at which the decision changes: the break-even premium per lamp
Honest report	Honest report	Result + range + top assumption + what would change the conclusion

[Source: Lecture07; Lab7]

5 5. Formulalar — a summary reference of the course formulas

This section collects all the formulas and calculation relations of weeks 5–7 — from the lectures, the labs and the templates. Notation of the bus model: C — battery capacity [kWh]; u — the usable share of the battery (unitless); e — energy consumption [kWh/km]; d — distance [km]; R — range [km]; E

- trip energy [kWh]. The rules for units, rounding and checks are in section 6; all values are illustrative.

5.1 5.1. Week 5 — the electric bus model

Trip energy. Consumption per kilometre multiplied by the distance.
$E = e \times d$ [kWh] = [kWh/km] × [km]
Range on a single charge. The battery's usable energy divided by the consumption.
$R = (u \times C) \div e \to 160$ [kWh] ÷ 1,2 [kWh/km] = 133 [km]
A unit check after Kosky. Units cancel like fractions.
$(u \times C)$ [kWh] ÷ e [kWh/km] = R [km]
The route's daily requirement. Laps multiplied by the lap length.
daily requirement = 8×12 [km] = 96 [km]
The template for writing an assumption.
We assume [assumption] because [reason]. If it is wrong, [effect on the result].
Energy balance (the campus electricity supply model).
supply = demand + losses

5.2 5.2. Week 6 — demand, capacity, flows, estimates

Utilisation. The same units and the same time basis on both sides.
Utilisation = Demand ÷ Capacity
Utilisation thresholds (a rule of thumb).
< ~{}0,8 — comfortable; ~{}0,8–1,0 — tight; $> 1,0$ — overload (a queue or refusals)
Power × time = energy.
Power × time = energy: 60 [kW] ×2 [h] = 120 [kWh]
Stock and flow.
change in stock = inflow − outflow
Buffer filling time.
filling time = free space ÷ (inflow − outflow) = 800 ÷ 40 = 20 [h]
Flow rule 1.
Flow = speed × density: 15 [km/h] ×20 [cars/km] = 300 [cars/h]
Flow rule 2.
number inside = arrivals × time inside: 10 [students/min] ×3 [min] = 30
The daily capacity of a stage.
capacity [units/day] = rate [units/h] × time [h/day] = $680 \times 12 = 8160$
Charging time.
time = energy ÷ power: 115 [kWh] ÷ 60 [kW] ≈ $1,9$ [h]
Average power over a window.
average power = energy ÷ window: 690 [kWh] ÷ 8 [h] ≈ 86 [kW]
Installation size from an estimate.
installation = daily energy ÷ yield per kW per day = $115 \div 4 \approx 29 \to$ ~{}30 [kW]; area =
30×5 [m²/kW] ≈ 150 [m²]
Good output with losses by stage.

output = input $\times(1-l1)\times(1-l2)\times(1-l3) = 7200\times 0,98\times 0,97\times 0,96 \approx 6571$ [units/day]

The capacity required for a target utilisation.

required capacity = peak demand ÷ target utilisation $= 7000 \div 0,85 \approx 8235$ [units/day]
≈
686 [units/h]

5.3 5.3. Week 7 — ranges, metrics, reporting

The mean of a small sample.

mean $= (x1 + x2 + \ldots + xn) \div n = (1,1 + 1,3 + 1,6) \div 3 = 1,33$ [kWh/km]

The median of a small sample.

median = the middle element in order $= 1,3$ [kWh/km] $(n = 3)$

The range of a result. Compute the result for every value of the range, not for one.

$R = 160$ [kWh] ÷ e: summer $1, \frac{1}{1}, \frac{3}{1}, 6 \to \frac{145}{123}/100$ [km]; winter $1, \frac{5}{1}, \frac{8}{2}, 0 \to \frac{107}{89}/80$
[km]

A share (percentage) — always of a stated base.

share = part ÷ base: 12 ÷ $20 = 60$% (corridor lighting); 12 ÷ $180 \approx 7$% (the building's peak)

Annual energy saving (the LED case).

saving [kWh/year] = saving per lamp [W] × lamps × hours [h/day] $\times 365$ ÷ 1 000

Triangulation (the faculty building).

8 000 [m²] $\times 0,35 \approx 2800$; 94 [kW] $\times 24$ [h] ≈ 2260; 70 000 ÷ $30 \approx 2330$ [kWh/day]

The break-even premium per lamp (the flip point).

break-even ≈ the annual saving per lamp × years of service: 25,8 [kWh] $\times 850$ [UZS] ≈ 21900
[UZS/year]

The covered share (the solar roof).

share = the installation's energy ÷ the building's energy $= 90$–$150 \div 2260 \approx 4$–7%

5.4 5.4. Units and conversions (weeks 6–7)

Powers-of-a-thousand prefixes.

1 kW $= 1000W$; $1MW = 1000$ kW; 1 kWh $= 1000Wh$

Megawatts to kilowatts.

$0,2MW = 200$ kW $(MW \to kWisa$ factor of 1 000)

Minutes to hours.

36 min ÷ $60 = 0,6h$(minutes are not decimal fractions of an hour)

The precision rule. Precise is not the same as correct.

"about 30 kW", not "29,17 kW" — one–two significant figures for rough estimates

[Source: Lecture05–Lecture07; Lab5–Lab7; the sheets of weeks 5–7]

6 6. Qoidalar — all the course rules in one checklist

All the rules of weeks 5–7 in one place, grouped under four themes. The wording is from the lectures, the labs and the course templates; the formulas are in section 5.

6.1 6.1. Rules of problem framing

- Sort what you hear: fact, inference, speculation, opinion — only facts and good inferences can anchor

the statement.

- A problem statement names the need, not the solution: no device, no app, no

vehicle (the solution-free test).

- Variables are measurable quantities with units: performance (what we judge), design (the levers), context

(the limits).

- “Works well” is defined by the stakeholders — write the performance down in numbers.

- The statement template: [Stakeholder] needs [outcome] in [system], judged by [measure], within

[constraints].

- The working definition of performance: the performance of [the system] is judged by [variables with units and targets], through the eyes of [the stakeholders].

- Goals: must-have = demand, nice-to-have = wish; do not turn a

wish into a requirement.

- A model is built for one clear question: first the question, then the model.

- Every model keeps what matters, discards what does not, and serves the purpose-question; “no map

is the network”.

- “All models are wrong, but some are useful” (George Box): usefulness is decided by the question, not by

completeness.

6.2 6.2. Rules of modelling

- Four types of models — physical, graphical, mathematical, computational; real

projects mix several types.

- The anatomy of a model: inputs (we choose), parameters (we fix), relationships (the rules), outputs

(we predict) — each with units.

- A model is a machine: numbers go in, a rule is applied, numbers come out; every number needs its

units.

- The week 4 variables become roles: performance → output, design → input, context → parameter.

- Need–Know–How–Solve: How includes the assumptions you make in order to solve the problem;

Solve — compute, check the units, interpret.

- A unit check after Kosky: units cancel like fractions — $(u \times C)$ [kWh] ÷ e [kWh/km] $=$

[km].

- Three kinds of assumptions: simplifying (we ignore), data (we fix), boundary (we leave

outside).

- Write every assumption with the template: "We assume … because …. If it is wrong, …".

- An assumption that is written down can be checked; a hidden one cannot.

- Classify the assumptions: Safe — holds under normal conditions; Risky — may

fail and change the answer; Need Data — cannot be said yet.

- Stress-test the model: change one assumption and explain the effect on the output.

- Three checks: units, order of magnitude, a known value; ask "Is this a reasonable

answer?" every time.

- The cycle: Frame → Build → Check → Revise — repeat until it is good enough for

the decision.

- The model's verdict: Too Simple / Fit for Purpose / Too Detailed; the goal is fit for purpose: the simplest

model that answers the question well enough to decide.

- Finish with one revision and one additional need for data.

- Every model has a range in which its assumptions are valid: know it and state it.

6.3 6.3. Rules of quantitative reasoning

- Demand is what the system must deliver; capacity is the maximum it can deliver; both need

units and a time basis.

- A number without a time basis is as incomplete as a number without units.

- Compare like with like: one unit and one time basis on both sides before

dividing.

- Utilisation = demand ÷ capacity: below ~{}0,8 — comfortable; around 1 — tight; above 1 —

demand is turned away or a queue grows.

- Size for the peak, not for the average: the average hides the peak (a 94 kW average against a 180 kW peak at

13:00).

- kW is an intensity, kWh is an amount; power × time = energy.

- Carry the units in square brackets: 60 [kW] $\times 2$ [h] $= 120$ [kWh]; the units check both the method

and the answer.

- Change in stock = inflow − outflow; when positive, something is accumulating (a queue, a buffer,

a battery).

- The flow rules (flow = speed × density; number inside = arrivals × time) hold only under

steady-state conditions.

- The bottleneck is the stage with the smallest capacity; it sets the system's maximum practical

throughput.

- Estimate before calculating: identify the quantity and its units → break into factors → round

each factor to one digit and multiply → check the units and compare with a known value.

- An order-of-magnitude estimate is correct to within a factor of ~{}10 and often better: 21 barbershops

against 23 in the directory.

- Bound the estimate: low / typical / high; if the decision does not change across the range —

stop refining $(23\text{–}38kW \rightarrow$ still "a rooftop installation").

- Precise is not the same as correct: one–two significant figures; "about 30 kW", not "29,17 kW".

- Most estimation errors are errors of units or of counting: confusing kW and kWh, the forgotten peak,

double counting, false precision.

- When an answer looks strange, check the units and the time basis first $(0,2MW =$

200 kW; $36\,\text{min} = 0,6h)$.

6.4 6.4. Rules for working with incomplete data and making decisions

- Name the data problem first: too little, missing, noisy, out of date / unrepresentative, wrong

definition / unit — the fix depends on the type.

- Five sources of numbers: measure, look up, ask, estimate, analogy — with different levels of trust;

write the source and the confidence next to every number.

- Use ranges, not single numbers: recompute the main result for the low,

typical and high values.

- A range tells the reader how far to trust the answer: “about 100 to 145 km”.

- With a small sample, write n next to every number; do not claim more than n

points can show.

- One odd value moves the mean a lot, and the median almost not at all; choose the metric

deliberately.

- Triangulate: estimate an important quantity by at least two independent routes; agreement

builds trust, disagreement points to where to look.

- The same numbers support different readings, and all can be true; ask for

which decision this number is.

- The decision chooses the metric, not the other way round: the median — the typical case; the mean — the total

load; the 90th percentile — the unlucky; the maximum — the worst case.

- Always ask: a percentage of what? Different baselines and boundaries give different but true

stories (60% against 7%).

- A result above 100% is a hint that the baseline boundary is wrong; use one boundary

before and after the change, and state it.

- Misleading comparisons need no false number: different denominators (“per what?”),

cherry-picked days (“which days were dropped?”), a truncated axis (“where does the axis start?”).

- Four questions for any interpretation: for which decision? which metric? which base, boundary

and time basis? which assumption would flip the conclusion?

- An honest report: result + range + top assumption + what would change the conclusion.
- Low confidence goes on the page: hiding it does not remove it.
- Find the flip point: the threshold at which the decision changes (the break-even premium ≈ the annual

saving × years of service).

[Source: Lecture05–Lecture07; Lab5–Lab7; the templates of weeks 5–7]

7 7. Topshiriqlar — all the assignments of weeks 5–7

A summary list of all the assignments of weeks 5–7: the assessed lab team works, the template sheets, the lecture exercises and the independent study. Full statements and solutions are in sections 2.3, 3.3 and 4.3.

Wk	Assignment	Format and grading	Source
5	Lab 5 “Build & Check a Model”: ten steps on the Lab 4 scenario card	team work, 45 min, graded per step 3/2/1/0	Lab5ProblemFraming II.txt
5	Model Building & Assumption Testing Worksheet	the week 5 sheet, 10 items	Lab5ModelWorksheet Template.txt
5	Engineering Problem Framing — the cross-cutting template A–H	team work, 30–40 min	EngineeringProblem FramingBlanket Template.txt
5	Milestone 2: a problem-framing document $+a$ list of assumptions (≥ 8, each safe / risky / needs data)	a team deliverable	Lecture05, slide 32
6	Lab 6 “Quantify the System”: ten steps on the case	team work, 35–40 min, assessed	Lab6Quantitative ReasoningI.txt
6	Quantitative Reasoning Worksheet	the week 6 sheet, 10 items	Lab6Worksheet Template.txt
6	Lecture 6: “Demand or Capacity?”, “Can Ten Buses Charge?”, “Estimate It in Your Team”	in class, in pairs, with answer keys	Lecture06, slides 11, 20, 27
6	Independent Study: Numbers for Your System (4 items)	not handed in — the team’s working material	Lecture06, slide 31
7	Lab 7 “Decide With Incomplete Data”: parts A–I	team work, 35–40 min, assessed	Lab7Quantitative ReasoningII.txt
7	Group Work Template: Engineering Decisions with Incomplete Data	the week 7 sheet, parts A–I	Lab7Engineering DecisionsIncompleteData Template.txt
7	Lecture 7: “Name the Problem”, “Read It Twice”, “Winter Range Explorer”, “Rooftop Solar”	in class and interactive, with keys	Lecture07, slides 11, 19, 27–28
7	Independent Study: Incomplete Data in Your System (4 items)	not handed in — the team’s working material	Lecture07, slide 31

Table 7.1. All the assignments of weeks 5–7

Project milestones. Week 5 ends with Milestone 2 — a problem-framing document and an explicit list of modelling assumptions (at least eight: what we assume, why, and what changes if it is wrong; each labelled safe / risky / needs data). By the course milestone plan (the weeks 1–4 guide), week 7 is the first quantitative pass over the model and a short in-class pitch (Milestone 3); lecture 7 closes with the words “working material for your system, ready for the second half of the course”.

The independent studies of weeks 6 and 7 are not handed in — they are the team’s material for its own system.

Grading of the labs. Lab 5: ten steps, each on the scale 3 (good/fully) / 2 (partially) / 1 (almost nothing) / 0 (missing); the work is graded live, and the steps are done in order. Lab 6 and Lab 7: group work in ten steps and nine parts A–I respectively, followed by the model answer walkthrough — EV Charging & Battery Test Station (Lab 5), Semiconductor Wafer Packaging Line (Lab 6), LED Lamp Reliability and

Energy-Saving Decision (Lab 7).
[Source: Lab5–Lab7; Lecture05–Lecture07; the milestone plan — the weeks 1–4 guide]

8 8. Cheat sheet: key facts and numbers

8.1 8.1. Key facts

- A model $= a$ simplified representation of a real system, built to answer a specific

question; “no map is the network”.

- Four types of models: physical (a blade in a wind tunnel), graphical (a one-line diagram),

mathematical (Ohm’s law), computational (a table of hourly demand).

- Four parts of a model: inputs, parameters, relationships, outputs — all with units; performance $\rightarrow$

output, design $\rightarrow$ input, context $\rightarrow$ parameter.

- The bus: $u \times C = 0,8 \times 200 = 160$ kWh; $R = 160 \div 1,2 = 133$ km; the daily requirement 96 km (8

laps $\times 12 km$); a margin of about 37 km.

- The range by assumptions: 133 (base) / 114 (full load) / 107 (old battery) / 89 (winter) / 71

(winter + old battery) km; the first three cases cover the 96 km.

- Solar: naively 5 kW $\times 24h = 120$ kWh/day, clear ≈ 31 kWh — nearly 4 times too high; kW is

power, kWh is energy.

- Utilisation: the faculty building 180 $\div 250 = 0,72$(comfortable); the canteen 360 $\div 400 = 0,90$(tight);

the bus 300 $\div 240 = 1,25$(overload); the depot 360 $\div 200 = 1,80$(overload).

- The average hides the peak: 94 kW average, 180 kW peak at 13:00, the transformer 250 kW — size for

the peak.

- Charging: 115 kWh $\div$ 60 kW $\approx 1,9 hper$ bus; six buses $= 690$ kWh over an 8 h window $\rightarrow$ an average

86 kW; in two batches $3 \times 60 = 180$ kW (utilisation 0,9), the time $2 \times 1,9 \approx 3,8h$.

- Ten buses: 1 150 kWh; ≈ 144 kW average; 3 chargers at a time; 4 batches; $4 \times 1,9 \approx$

7,7 h — it fits almost without margin.

- The solar estimate for a bus: 115 ÷ $4 \approx 29 \rightarrow$ about 30 kW on about 150 m²; the bounds of the estimate 23–38

kW.

- The semiconductor line: capacities 8 160 / 7 440 / 6 960 / 7 200 / 7 680 units/day; the bottleneck is

Wire Bonding 6 960; at the peak of 7 000 its utilisation is 1,01.

- Line losses: $7200 \times 0,98 \times 0,97 \times 0,96 \approx 6571$ good units $(\approx 629 of 7200$, about 9% never shipped);

the buffer: $600 - 560 = 40$ units/h, full in 20 h.

- The LED case: a saving of 124–216 MWh/year (typically 167), 87–216 million UZS/year; the median lamp

power $12,1W(n = 5)$; the break-even premium $\approx 200000 UZSat 40000h$.

8.2 8.2. Key quotes and definitions

- “All models are wrong, but some are useful” —

George Box, statistician.

- “A model is a machine: numbers go in, a rule is applied, numbers come

out”.

- “Every number needs its units”.

- “Neither map is the grid”: each is a model for one type of

question.

- “Choose the simplest model that answers the question well enough to

decide”.

- “An assumption that is written down can be checked. A hidden one

cannot”.

- “Is this a reasonable answer?” — Kosky’s question for every engineering

result.

- “Size for the peak, not for the average”.

- “Energy was never the problem. Power

was”.

- “Precise is not the same as accurate”.

- “The decision picks the metric, not the other way

round”.

- “A good report gives the answer, how far to trust it, and what would

change it”.

8.3 8.3. Ten one-line answers

- A model = a simplification for the sake of one question: the question decides what to keep and what to throw away.

- Types of models: physical, graphical, mathematical, computational; projects

mix them.

- Anatomy: inputs → parameters → relationships → outputs, all with units.

- Assumptions: simplifying (we ignore), data (we fix), boundary (we leave outside) — and

each one with the template.

- Fit for purpose = the simplest model sufficient for the decision; too simple and too

detailed are the two failures.

- Checks: units, order of magnitude, a known value — “Is this a reasonable answer?”.

- Utilisation = demand ÷ capacity; $<$ ~{}0,8 comfortable, >1 — a queue; size for the peak.

- kW is a rate, kWh is an amount; power × time = energy; units in square

brackets.

- Estimation: break into factors, round to one digit, check the units, bound with

a range.

- Incomplete data: name the problem type, range $+n$, triangulate; the report — the result,

the range, the top assumption, what would change the conclusion.

8.4 8.4. What comes next

Week 8 — the Midterm Assessment: a written, paper-based test covering the first seven weeks $(30\% of the$ grade, ILO 1–4 — all the logic of this guide). By the course description, after quantitative reasoning the semester moves on to decision-making under trade-offs (efficiency, cost, fairness, accessibility — ILO-4) and to system dynamics and simple simulations under uncertainty (delays, queues, throughput, bottlenecks — ILO-3): lecture 6 announces queues in week 9 and trade-offs in week 10, lecture 5 — sensitivity analysis in week 11; by the milestone plan, week 10 hands in the trade-off analysis of two alternative designs (Milestone 4), and week 12 — the draft of a scenario simulation (Milestone 5). Before the Midterm, revise: a solution-free problem statement and variables with units; the four types and four parts of a model; the three kinds of assumptions and the template for writing them; fit for purpose and the three checks; demand, capacity, utilisation and peaks; intensities, flows and stocks; order of magnitude and bounding with a range; the five types of incomplete data, triangulation and the honest report — plus weeks 1–4 (ethics and the NSPE codes, systems and boundaries, PFD, evidence and D/W specifications).

[Source: Lecture07, slide 32 (Midterm); the previews in Lecture05–Lecture06; the milestone plan — the weeks 1–4 guide]